2025 Chainsaw Manual
(reduced, English Only)
-David Malawey

STIHL

STIHL MS 250

Instruction Manual
Manual de instrucciones

Displacement: 45.4cc
Oil capacity: 200cc
Chain: 0.325" RM3

WARNING To reduce the risk of kickback injury use STIHL reduced kickback bar and STIHL low kickback chain as specified in this manual or other available low kickback components.

ADVERTENCIA Para reducir el riesgo de lesionarse como resultado de un culatazo, utilice la barra y la cadena de contragolpe reducido de la forma especificada en este manual o de otros componentes reductores de contragolpe.

Read Instruction Manual thoroughly before use and follow all safety precautions – improper use can cause serious or fatal injury.

Antes de usar la máquina lea y siga todas las precauciones de seguridad dadas en el manual de instrucciones – el uso incorrecto puede causar lesiones graves o mortales.

Instruction Manual
1 - 60

Manual de instrucciones
61 - 126

Displacement:

Contents

Guide to Using this Manual 2
Safety Precautions and Working Techniques 3
Cutting Attachment 27
Mounting the Bar and Chain 28
Tensioning the Chain 29
Checking Chain Tension 29
Fuel 29
Fueling 31
Chain Lubricant 33
Filling Chain Oil Tank 34
Checking Chain Lubrication 34
Chain Brake 35
Winter Operation 36
Starting / Stopping the Engine 36
Operating Instructions 40
Taking Care of the Guide Bar 41
Air Filter System 42
Cleaning the Air Filter 42
Engine Management 43
Adjusting the Carburetor 43
Spark Plug 44
Engine Running Behavior 45
Storing the Machine 46
Checking and Replacing the Chain Sprocket 46
Maintaining and Sharpening the Saw Chain 47
Maintenance and Care 51
Main Parts 53
Specifications 55
Ordering Spare Parts 56
Maintenance and Repairs 56
Disposal 56
Limited Warranty 57
STIHL Incorporated Federal Emission Control Warranty Statement 57
Trademarks 59

Allow only persons who fully understand this manual to operate your chain saw.

To receive maximum performance and satisfaction from your STIHL chain saw, it is important that you read, understand and follow the safety precautions and the operating and maintenance instructions in chapter "Safety Precautions and Working Techniques" before using your chain saw. For further information you can go to www.stihlusa.com.

Contact your STIHL dealer or the STIHL distributor for your area if you do not understand any of the instructions in this manual.

⚠ WARNING

Because a chain saw is a high-speed wood-cutting tool, some special safety precautions must be observed as with any other power saw to reduce the risk of personal injury. Careless or improper use may cause serious or even fatal injury.

Original Instruction Manual
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STIHL

Guide to Using this Manual

This Instruction Manual refers to a STIHL chain saw, also called a machine in this Instruction Manual.

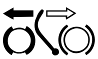
Pictograms

The meanings of the pictograms attached to or embossed on the machine are explained in this manual.

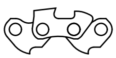
Depending on the model concerned, the following pictograms may be on your machine.

Fuel tank; fuel mixture of gasoline and engine oil

Chain oil tank; chain oil

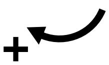


Engaging and disengaging the STIHL Quickstop chain brake



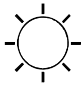
Direction of chain rotation

Ematic; chain oil quantity control



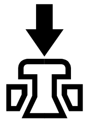
Tension the chain

Intake air preheating for winter operation

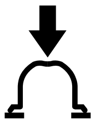


Intake air for summer operation

Handle heating



Operate decompression valve



Operate manual fuel pump

Symbols in Text

Many operating and safety instructions are supported by illustrations.

The individual steps or procedures described in the manual may be shown in different ways:

- A bullet indicates a step or procedure.

A description of a step or procedure that refers directly to an illustration may contain item numbers that appear in the illustration. For example:

- Remove the screw (1)
- Pull the spark arresting screen (2) upwards out of the muffler

In addition to the operating instructions, this manual may contain paragraphs that require your special attention. Such paragraphs are indicated with the symbols and signal words described below:

DANGER

Indicates a hazardous situation that, if not avoided, will result in death or serious injury.

WARNING

Indicates a hazardous situation that, if not avoided, could result in death or serious injury.

NOTICE

Indicates a risk of property damage, including damage to the machine or its individual components.

Engineering Improvements

STIHL’s philosophy is to continually improve all of its products. As a result, engineering changes and improvements are made from time to time. Therefore, some changes, modifications and improvements may not be covered in this manual. If the operating characteristics or the appearance of your machine differs from those described in this manual, please contact your STIHL dealer or the STIHL distributor for your area for assistance.

Safety Precautions and Working Techniques

Because a chain saw is a high-speed, fast-cutting power tool, special safety precautions must be observed to reduce the risk of personal injury.

It is important that you read, fully understand and observe the following safety precautions and warnings. Read the instruction manual and the safety instructions periodically. Careless or improper use may cause serious or fatal injury. Save the instruction manual for future reference.

WARNING

The use of this chain saw may be hazardous. The saw chain has many sharp cutters. If the cutters contact your flesh, they will cut you, even if the chain is not moving.

WARNING

Reactive forces, including kickback, can be dangerous. Pay special attention to the section on reactive forces.

Have your STIHL dealer show you how to operate your chain saw. All safety precautions that are generally observed when working with an axe or a hand saw also apply to the operation of chain saws. Observe all applicable federal, state and local safety regulations, standards and ordinances. When using a chain saw for logging purposes, for instance, refer to the OSHA regulations for "logging operations" at 29 Code of Federal Regulations 1910.266.

WARNING

Do not lend or rent your chain saw without the instruction manual. Be sure that anyone using it understands the information contained in this manual.

The use of noise emitting chain saws may be restricted to certain times by national, state or local regulations.

Use your chain saw only for cutting wooden objects.

WARNING

Do not use it for other purposes, since misuse may result in personal injury or property damage, including damage to the chain saw.

WARNING

Minors should never be allowed to use this chain saw. Bystanders, especially children, and animals should not be allowed in the area where it is in use.

Most of these safety precautions and warnings apply to the use of all STIHL chain saws. Different models may have different parts and controls. See the appropriate section of your instruction manual for a description of the controls and the function of the parts of your model.

WARNING

Always stop the engine and activate the QuickStop Chainbrake before transporting or carrying out any work on the chain saw. This avoids the risk of the engine starting unintentionally.

STIHL recommends the use of genuine STIHL replacement parts. They are specifically designed to match your model and meet your performance requirements.

Safe use of a chain saw involves

1 the operator

2 the chain saw

3 the use of the chain saw.

THE OPERATOR

Physical Condition

You must be in good physical condition and mental health and not under the influence of any substance (drugs, alcohol, etc.) which might impair vision, dexterity or judgment. Do not operate this chain saw when you are fatigued.

WARNING

Be alert – if you get tired, take a break. Tiredness may result in loss of control. Working with any power tool can be strenuous. If you have any condition that might be aggravated by strenuous work, check with your doctor before operating this chain saw.

WARNING

Prolonged use of a chain saw (or other power tools) exposing the operator to vibrations may produce whitefinger disease (Raynaud's phenomenon) or carpal tunnel syndrome.

These conditions reduce the hand's ability to feel and regulate temperature, produce numbness and burning sensations and may cause nerve and circulation damage and tissue necrosis.

All factors which contribute to white-finger disease are not known, but cold weather, smoking and diseases or physical conditions that affect blood vessels and blood transport, as well as high vibration levels and long periods of exposure to vibration are mentioned as factors in the development of whitefinger disease. In order to reduce the risk of whitefinger disease and carpal tunnel syndrome, please note the following:

– Most STIHL chain saws are available with an anti-vibration ("AV") system designed to reduce the transmission of vibrations created by the chain saw to the operator's hands. An AV system is recommended for those persons using chain saws on a regular or sustained basis.
– Wear gloves and keep your hands warm. Heated handles, which are available on some STIHL chain saws, are recommended for cold weather use.
– Keep the AV system well maintained. A chain saw with loose components or with damaged or worn AV elements will tend to have higher vibration levels.
– Keep the saw chain sharp and well maintained. A dull saw chain will increase cutting time, and pressing a dull saw chain through wood will increase the vibrations transmitted to your hands.
– Maintain a firm grip at all times, but do not squeeze the handles with constant, excessive pressure. Take frequent breaks.

All the above-mentioned precautions do not guarantee that you will not sustain whitefinger disease or carpal tunnel syndrome. Therefore, continual and regular users should closely monitor the condition of their hands and fingers. If any of the above symptoms appear, seek medical advice immediately.

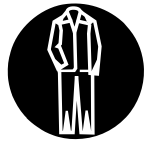
WARNING

The ignition system of the STIHL unit produces an electromagnetic field of a very low intensity. This field may interfere with some pacemakers. To reduce the risk of serious or fatal injury, persons with a pacemaker should consult their physician and the pacemaker manufacturer before operating this chain saw.

Proper Clothing

WARNING

To reduce the risk of injury, the operator should wear proper protective apparel.



Clothing must be sturdy and snug-fitting, but allow complete freedom of movement.To reduce the risk of cut injuries, wear the type of overalls, long pants or chaps that contain pads of cut-retardant material. Avoid loose-fitting jackets, scarfs, neckties, jewelry, flared or cuffed pants, unconfined long hair or anything that could become caught on branches, brush or the moving parts of the chain saw. Secure hair so it is above shoulder level..

Good footing is very important. Wear sturdy boots with nonslip soles. Steel-toed safety boots are recommended. Never wear sandals, flip-flops or go barefoot.

Always wear heavy-duty work gloves (e.g. made of leather or wear resistant material) when handling the chain saw and the cutting tool. Heavy-duty, nonslip gloves improve your grip and help to protect your hands.

To reduce the risk of injury to your eyes never operate your power tool unless wearing goggles or properly fitted protective glasses with adequate top and side protection complying with ANSI Z87 "+" (or your applicable national standard). If there is a risk of injury to your face, STIHL recommends that you also wear a face shield or face screen over your goggles or protective glasses.

Wear an approved safety hard hat to reduce the risk of injury to your head. Chain saw noise may damage your hearing. Wear sound barriers (ear plugs or ear mufflers) to help protect your hearing. Continual and regular users should have their hearing checked regularly.

Be particularly alert and cautious when wearing hearing protection because your ability to hear warnings (shouts, alarms, etc.) is restricted.

THE CHAIN SAW

For illustrations and definitions of the chain saw parts see the chapter on "Main Parts."

WARNING

Never modify this chain saw in any way. Only attachments supplied by STIHL or expressly approved by STIHL for use with the specific STIHL model are authorized. Although certain unauthorized attachments are useable with STIHL chain saws, their use may, in fact, be extremely dangerous.

WARNING

Never operate your chain saw if it is damaged, improperly adjusted or maintained, or not completely and securely assembled.

If this chain saw is subjected to unusually high loads for which it was not designed (e.g. heavy impact or a fall), always check that it is in good condition before continuing work. Check in particular that the fuel system is tight (no leaks) and that the controls and safety devices are working properly. Do not continue operating this chain saw if it is damaged. In case of doubt, have it checked by your STIHL servicing dealer.

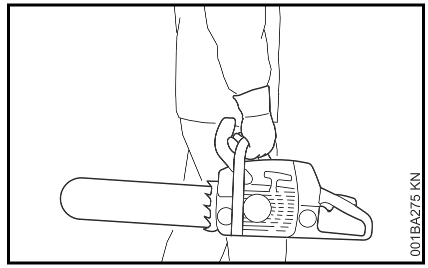
THE USE OF THE CHAIN SAW

Transporting the Chain Saw

WARNING

Always stop the engine before putting the chain saw down. Carrying a chain saw with the engine running may be extremely dangerous.

Accidental acceleration of the engine can cause the saw chain to rotate. Always engage the chain brake when taking more than a few steps.



By hand: When transporting your chain saw by hand, the engine must be shut off and the chain saw must be in the proper position, i.e., grip the top handle and place the muffler away from the body; the chain guard (scabbard) should be over the saw chain and guide bar, which should point backwards, away from the direction in which you are walking.

By vehicle: When transporting in a vehicle, keep saw chain and bar covered with the chain guard (scabbard). Properly secure your chain saw to prevent turnover, fuel spillage and damage to the chain saw.

Fuel

Your STIHL chain saw uses an oil-gasoline mixture for fuel (see the "Fuel" chapter in this instruction manual).

WARNING

Gasoline is an extremely flammable fuel. If spilled and ignited by a spark or other ignition source, it can cause fire and serious burn injury or property damage. Use extreme caution when handling gasoline or fuel mix. Do not smoke or bring any fire or flame near the fuel or the chain saw. Note that combustible fuel vapor may escape from the fuel system.

Fueling Instructions

WARNING

Pick a Safe Location

To reduce the risk of fire and explosion, fuel your chain saw in a well-ventillated area, outdoors away from flames, pilot lights, heaters, electric motors, and other sources of ignition. Vapors can be ignited by a spark or flame many feet away. Select bare ground for fueling and move at least 10 feet (3 m) from the fueling spot before starting the engine. Wipe off any spilled fuel before starting your chain saw. Take care not to get fuel on your clothing. If this happens, change your clothing immediately.

Allow the Saw to Cool Before Removing the Fuel Cap

WARNING

Gasoline vapor pressure may build up inside the fuel tank. The amount of pressure depends on a number of factors such as the fuel used, altitude and temperature. To reduce the risk of burns and other personal injury from escaping gas, vapor and fumes, always shut off the engine and allow it to cool before removing the fuel cap.

The engine is air cooled. When it is shut off, cooling air is no longer drawn across the cylinder and engine temperatures will rise for several minutes before starting to cool. In hot environments, cooling will take longer. To reduce the risk of burns and other personal injury from escaping gas, vapor and fumes, allow the saw to cool. If you need to refuel before completing a job, turn off the machine and allow the engine to cool before opening the fuel tank.

Fuel Spraying or “Geysering”

WARNING

Removing the cap on a pressurized fuel tank can result in gasoline, vapors and fumes being forcefully sprayed out from the fuel tank in all directions. The escaping gasoline, vapors or fumes can cause serious personal injury, including fire and burn injury, or property damage.

Sometimes also referred to as "fuel geysering," fuel spraying is an expulsion of fuel, vapors and fumes which can occur in hot conditions, or when the engine is hot, and the tank is opened without allowing the saw to cool adequately. It is more likely to occur when the fuel tank is half full or more.

Pressure is caused by fuel and heat and can occur even if the engine has not been running. When gasoline in the fuel tank is heated (by ambient temperatures, heat from the engine, or other sources), vapor pressure will increase inside the fuel tank.

Some blends of gasoline, particularly those designed for use in winter, are more volatile and may cause tanks to pressurize more quickly or create greater pressure. At higher altitudes, fuel tank pressurization is more likely.

How to Avoid Fuel Spraying

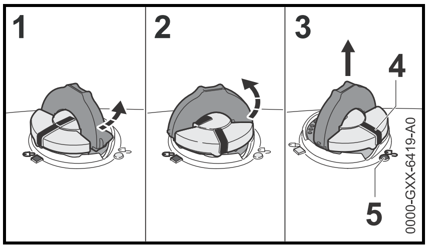
Removing the fuel cap on a pressurized tank can result in gasoline, vapors and fumes being forcefully sprayed out from the fuel tank in all directions. To reduce the risk of burns, serious injuries or property damage from fuel spraying:

- Follow the fueling instructions in this chapter.
- Always assume your fuel tank is pressurized.
- Allow the chain saw to cool before removing the fuel cap.
- In hot environments, cooling will take longer.
- The engine is air cooled. When it is shut off, cooling air is no longer drawn across the cylinder and the engine temperature will rise for several minues before starting to cool.

After the saw has cooled appropriately, follow the safety instructions in this chapter for removing the cap. Never remove the cap by turning it directly to the open position. First check for residual pressure by turning the cap slowly to the vent position, approximately 1/8 turn counter-clockwise. Use only good quality fuel that is appropriate for the season (summer v. winter blends). Some blends of gasoline, particularly winter blends, are more volatile and can contribute to fuel spraying.

Removing the Toolless Fuel Cap: Turn Slowly and Stop in the Vent Position

⚠ WARNING

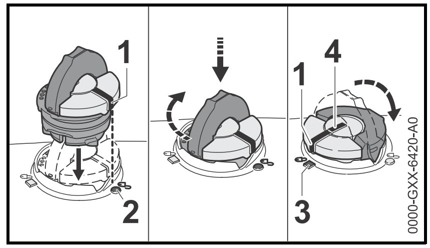


After allowing the chain saw to cool, remove the fuel filler cap slowly and carefully to allow any remaining pressure build-up in the tank to release:

– Flip up the grip and press the cap down firmly (1).
– While maintaining steady, downward pressure, turn the cap slowly counter-clockwise to the vent position (2), approximately a 1/8 turn of the cap.
– If any significant venting occurs, immediately re-seal the tank by turning the cap clockwise to the closed position. Allow the saw to cool further before attempting to open the tank.
– Turn the cap to the open position (3) only after the contents of the tank are no longer under pressure. In the open position, the exterior positioning mark (4) on the cap will line up with the "unlocked" symbol (5) on the fuel tank housing.
– Never remove the cap by turning it directly to the open position. First allow the saw to cool adequately and then release any residual pressure at the vent position (2).
– Never attempt to remove the cap while the engine is still hot or running.

Installing the Toolless Fuel Cap

⚠ WARNING

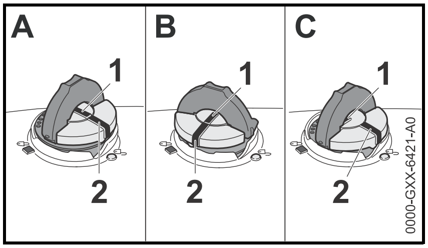


An improperly tightened fuel filler cap can loosen or come off and spill quantities of fuel. To reduce the risk of fuel spillage and fire from an improperly installed fuel cap, correctly position and tighten the cap in the fuel tank opening:

– Raise the grip on the top of the cap until it is upright at a 90° angle. Insert the cap in the fuel tank opening with the exterior positioning mark (1) lined up with the "unlocked" symbol (2) on the fuel tank housing.
– Using the grip, press the cap down firmly while turning it clockwise to the closed position (approximately 1/4 urn). In the closed position, the interior (4) and exterior (1) positioning marks will align with the "locked" symbol (3) on the fuel tank housing.
– Fold the grip flush with the top of the cap and check for tightness.

Misaligned, Damaged or Broken Cap

WARNING

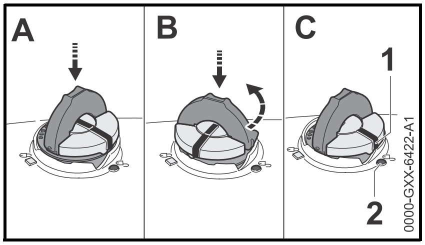


If the cap does not drop fully into the fuel tank opening when the positioning marks (1, 2) line up, or if it does not tighten properly when turned, the base of the cap may be prematurely rotated in relation to the top. Such misalignment can result from handling, cleaning or an improper attempt at tightening.

– Illustrations A and B: The base of the cap is prematurely rotated to the closed position and is not in the correct starting position for installation. The tank will not seal in this configuration. Note: in Illustrations A and B, the interior positioning marks (1) are in line with the exterior position marks (2).

– Illustration C: The bottom of the cap is in the correct starting position for installation. Note: In Illustration C, the interior positioning mark (1) is under the grip and not in line with the outer position mark (2).

To return the base of the cap to the proper starting position for installation:



- Drop the cap into the fuel tank opening (A).
- Next, turn the cap counter-clockwise with slight pressure until it drops fully into the fuel tank opening (approximately 1/4 turn) (B). This will rotate the base of the cap into the correct starting position for installation (C). The exterior positioning mark (1) on the cap will line up with the "unlocked" symbol (2) on the fuel tank housing. The interior positioning mark should be under the grip and not in line with the outer positioning mark (1).
- Then, turn the cap clockwise, closing it normally.

If your fuel cap still does not tighten properly, it may be damaged or broken. Stop using the chain saw and take it to your authorized STIHL dealer for repair.

Vapor Lock

WARNING

Vapor lock occurs when fuel in the fuel line or carburetor vaporizes, causing bubbles to block the free flow of liquid fuel into the carburetor. Vapor lock cannot be relieved or affected by opening the fuel tank. Removing the fuel filler cap without first allowing the chain saw to cool adequately can result in fuel spraying. Always follow the instructions in this section when removing the fuel cap.

To relieve vapor lock:

– Place the Master Control Lever in the cold start position and pull the starter cord approximately 20 times to clear the vapor and send liquid fuel into the carburetor.

– To start the chain saw, move the Master Control Lever to the starting throttle position and pull the starter cord approximately 10 times.

– If your chain saw will not restart, or if vapor lock occurs again, the chain saw is being used in conditions too extreme for the fuel being used. Discontinue use and let the engine cool completely before attempting to start the chain saw.

Before Operation

Take off the chain guard (scabbard) and inspect the chain saw for proper condition and operation. (See the maintenance chart near the end of the instruction manual.)

WARNING

Always check your chain saw for proper condition and operation before starting, particularly the throttle trigger, throttle trigger lockout, stop switch and cutting attachment. The throttle trigger must move freely and always spring back to the idle position. The Master Control Lever / stop switch must move easily to **STOP**, **0** or . Never attempt to modify the controls or safety devices.

WARNING

Check fuel system for leaks, especially the visible parts, e.g., filler cap, hose connections, manual fuel pump (only for chain saws equipped with a manual fuel pump). Do not start the engine if there are leaks or damage – risk of fire. Have the chain saw repaired by a STIHL servicing dealer before using it.

WARNING

Check that the spark plug boot is securely mounted on the spark plug – a loose boot may cause arcing that could ignite combustible fumes and cause a fire.

For proper assembly of the bar and saw chain follow the procedure described in the chapter "Mounting the Bar and Chain" of your instruction manual. STIHL Oilomatic saw chain, guide bar and sprocket must match each other in gauge and pitch. Before replacing any bar and chain, see the chapter entitled "Specifications" in the instruction manual and the chapter "Reactive Forces including Kickback".

Since longer bars add weight and may be more difficult to control, select the shortest bar that will meet your cutting needs.

WARNING

Proper tension of the chain is extremely important. In order to avoid improper setting, the tensioning procedure must be followed as described in your manual. Always make sure the hexagonal nut(s) for the sprocket cover is (are) tightened securely after tensioning the saw chain in order to secure the bar. Never start the chain saw with the sprocket cover loose. Check chain tension once more after having tightened the nut(s) and thereafter at regular intervals (whenever the saw is shut off). If the saw chain becomes loose while cutting, shut off the engine and then tighten. Never try to adjust the saw chain while the engine is running.

WARNING

After adjusting a saw chain, start the chain saw, let the engine run for a while, then switch engine off and recheck saw chain tension. Proper saw chain tension is very important at all times.

Keep the handles clean and dry at all times; it is particularly important to keep them free of moisture, pitch, oil, fuel mix, grease or resin in order for you to maintain a firm grip and properly control your chain saw .

WARNING

Be sure that the guide bar and saw chain are clear of you and all other obstructions and objects, including the ground. If the upper quadrant of the tip of the bar touches any object, it may cause kickback to occur (see section on reactive forces). Never attempt to start the chain saw when the guide bar is in a cut or kerf.

For specific starting instructions, see the appropriate section of your instruction manual.

Starting

WARNING

To reduce the risk of fire and burn injuries, start the engine at least 10 feet (3 m) from the fueling spot, outdoors only.

Start and operate your chain saw without assistance. For specific starting instructions, see the appropriate section of the instruction manual. Proper starting methods reduce the risk of injury.

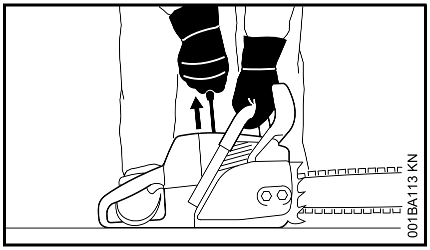
WARNING

To reduce the risk of injury from saw chain contact and / or reactive forces, the chain brake must be engaged when starting the chain saw. If your chain saw is equipped with the Quickstop Plus chain brake system, it is not sufficient to engage that brake only for starting, because the saw chain may begin to rotate at high speed when the throttle trigger lockout is depressed (releasing the brake) in order to blip the throttle trigger after starting.

WARNING

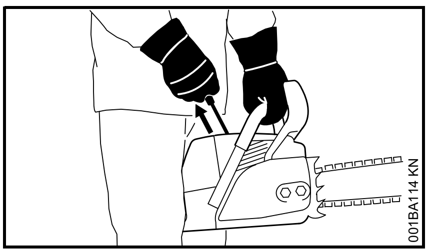
Do not drop start. This method is very dangerous because you may lose control of the chain saw.

There are two recommended methods for starting your chain saw.



With the **first** recommended **method**, the chain saw is started on the ground. Make sure the chain brake is engaged (see "Chain Brake" chapter in your instruction manual) and place the chain saw on firm ground or other solid surface in an open area. Maintain good balance and secure footing.

Grip the front handlebar of the saw firmly with your left hand and press down. For saws with a rear handle level with the ground, put the toe of your right foot into the rear handle and press down. With your right hand pull out the starter grip slowly until you feel a definite resistance and then give it a brisk, strong pull.



The **second** recommended **method** for starting your chain saw allows you to start the saw without placing it on the ground. Make sure the chain brake is engaged, grip the front handle of the chain saw firmly with your left hand. Keep your arm on the front handle in a locked (straight) position. Hold the rear handle of the saw tightly between your legs just above the knees. Maintain good balance and secure footing. Pull the starting grip slowly with your right hand until you feel a definite resistance and then give it a brisk, strong pull.

WARNING

Be sure that the guide bar and saw chain are clear of you and all other obstructions and objects, including the ground. When the engine is started, the engine speed with the starting throttle lock engaged will be fast enough for the clutch to engage the sprocket and, if the chain brake is not activated, turn the saw chain. If the upper quadrant of the tip of the bar touches any object, it may cause kickback to occur (see section on reactive forces). To reduce this risk, always engage the chain brake before starting. Never attempt to start the chain saw when the guide bar is in a cut or kerf.

Once the engine has started, immediately blip the throttle trigger, which should release the Master Control lever to the run position and allow the engine to slow down to idle.

Always disengage chain brake before accelerating engine and before starting cutting work. The only exception to this rule is when you check operation of the chain brake. High revs with the chain brake engaged (chain locked) will quickly damage the powerhead and chain drive (clutch, chain brake).

WARNING

When you pull the starter grip, do not wrap the starter rope around your hand. Do not let the grip snap back, but guide the starter rope to rewind it properly. Failure to follow this procedure may result in injury to your hand or fingers and may damage the starter mechanism.

Important Adjustments

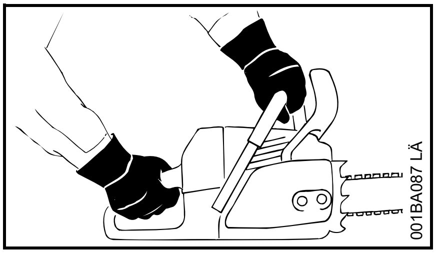
WARNING

To reduce the risk of personal injury from loss of control and / or contact with the running cutting tool, do not use your chain saw with incorrect idle adjustment. At correct idle speed, the cutting tool should not move. For directions on how to adjust idle speed, see the appropriate section of your instruction manual.

If you cannot set the correct idle speed, have your STIHL dealer check your chain saw and make proper adjustments and repairs.

Holding and Controlling the Chain Saw

Always hold the chain saw firmly with both hands when the engine is running. Place your left hand on the front handle bar and your right hand on the rear handle and throttle trigger.



Left-handers should follow these instructions too. Wrap your fingers tightly around the handles, keeping the handles cradled between your thumb and forefinger. With your hands in this position, you can best oppose and absorb the push, pull and kickback forces of your saw without losing control (see section on reactive forces).

WARNING

To reduce the risk of serious or fatal injury to the operator or bystanders from loss of control, never use the chain saw with one hand. It is more difficult for you to control reactive forces and to prevent the bar and chain from skating or bouncing along the limb or log.

WARNING

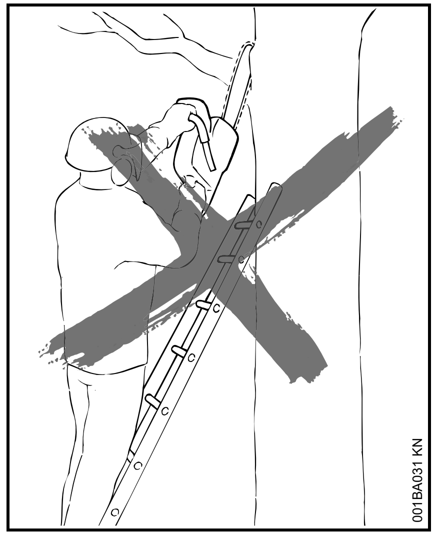
To reduce the risk of cut injuries, keep hands and feet away from the cutting tool. Never touch a moving cutting tool with your hand or any other part of your body.

WARNING

Keep proper footing and balance at all times. Special care must be taken in slippery conditions (wet ground, snow) and in difficult, overgrown terrain. Be extremely cautious when working on slopes or uneven ground. Watch for hidden obstacles such as tree stumps, roots, rocks, holes and ditches to avoid stumbling. There is increased danger of slipping on freshly debarked logs. For better footing, clear away fallen branches, scrub and cuttings. Use extreme caution when cutting small-size brush, branches and saplings because slender material may catch the saw chain and be whipped toward you or pull you off balance.

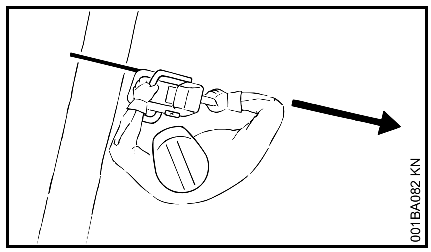
WARNING

Take extreme care in wet and freezing weather (rain, snow, ice). Put off the work when the weather is windy, stormy or rainfall is heavy.



WARNING

To reduce the risk of injury from loss of control, never work on a ladder or any other insecure support. Never hold the machine above shoulder height. Do not overreach.



Position the chain saw in such a way that your body is clear of the cutting attachment whenever the engine is running. Stand to the left of cut while bucking.

Never put pressure on the chain saw when reaching the end of a cut. The pressure may cause the bar and rotating saw chain to pop out of the cut or kerf, go out of control and strike the operator or some other object. If the rotating saw chain strikes some other object, a reactive force may cause the moving saw chain to strike the operator.

STIHL recommends that first-time users should cut logs on a sawhorse – see "Cutting small logs."

Working Conditions

Operate and start your chain saw only outdoors in a well-ventilated area. Operate it under good visibility and daylight conditions only. Work carefully.

WARNING

Your chain saw is a one-person machine. Do not allow other persons in the general work area, even when starting. Stop the engine immediately if you are approached.

WARNING

Even though bystanders should be kept away from the running chain saw, never work alone. Keep within calling distance of others in case help is needed.

WARNING

To reduce the risk of injury to bystanders and damage to property, never let your chain saw run unattended. When it is not in use (e. g. during a work break), shut it off and make sure that unauthorized persons do not use it.

WARNING

As soon as the engine is running, this product generates toxic exhaust fumes containing chemicals, such as unburned hydrocarbons (including benzene) and carbon monoxide, that are known to cause respiratory problems, cancer, birth defects, or other reproductive harm. Some of the gases (e. g. carbon monoxide) may be colorless and odorless. To reduce the risk of serious or fatal injury/illness from inhaling toxic fumes, never run the machine indoors or in poorly ventilated locations. If exhaust fumes become concentrated due to insufficient ventilation, clear obstructions from work area to permit proper ventilation before proceeding and/or take frequent breaks to allow fumes to dissipate before they become concentrated.

WARNING

Operate your chain saw so that it produces a minimum of noise and emissions – do not run engine unnecessarily and accelerate the engine only for cutting.

WARNING

Use of this chain saw (including sharpening the saw chain) can also generate dust, mist and fumes containing chemicals that are known to cause respiratory problems, cancer, birth defects, or other reproductive harm. If you are unfamiliar with the risks associated with the particular dust, mist or fume at issue, consult your employer, governmental agencies such as OSHA and NIOSH and other sources on hazardous materials. California and some other authorities, for instance, have published lists of substances known to cause cancer, reproductive toxicity, etc.

WARNING

Inhalation of certain dusts, especially organic dusts such as mold or pollen, can cause susceptible persons to have an allergic or asthmatic reaction. Substantial or repeated inhalation of dust and other airborne contaminants, in particular those with a smaller particle size, may cause respiratory or other illnesses. This includes wood dust, especially from hardwoods, but also from some softwoods such as Western Red Cedar. Control dust (such as saw dust), mists (such as oil mist from chain lubrication) and engine fumes at the source where possible. Use good work practices, such as always cutting with a properly sharpened saw chain (which produces wood chips rather than fine dust) and operating the unit so that the wind or operating process directs any dust raised by the chain saw away from the operator. Follow the recommendations of

EPA/OSHA/NIOSH and occupational and trade associations with respect to dust ("particulate matter"). When the inhalation of dust cannot be substantially controlled, i.e., kept at or near the ambient (background) level, the operator and any bystanders should wear a respirator approved by NIOSH / MSHA for the type of dust encountered.

WARNING

Breathing asbestos dust is dangerous and can cause severe or fatal injury, respiratory illness or cancer. The use and disposal of asbestos-containing products have been strictly regulated by OSHA and the Environmental Protection Agency. Do not use your chain saw to cut or disturb asbestos or asbestos-containing products. If you have any reason to believe that you might be cutting asbestos, immediately stop cutting and contact your employer or a local OSHA representative.

Operating Instructions

WARNING

Do not operate your chain saw with the starting throttle lock engaged. Cutting with the starting throttle lock engaged does not permit the operator proper control of the chain saw or saw chain speed. Begin and continue cutting with the saw at full throttle, engage the bumper spike firmly in the wood (if possible) and then continue cutting. Always work with the bumper spike so that you have better control of the saw. If you work without the bumper spike the chain saw may pull you forwards suddenly.

WARNING

Never touch a saw chain with your hand or any part of your body when the engine is running, even when the chain is not rotating.

In the event of an emergency, switch off the engine immediately – move the Master Control Lever to **STOP**, **0** or 0 .

WARNING

Always stop the engine before putting the chain saw down.

WARNING

The saw chain continues to move for a short period after the throttle trigger is released (flywheel effect).

Accelerating the engine while the saw chain is blocked increases the load and will cause the clutch to slip continuously. This may occur if the throttle is depressed for more than a few seconds when the saw chain is pinched in the cut or the chain brake is engaged. It can result in overheating and damage to important components (e. g. clutch, polymer housing components) – which can then increase the risk of injury, e. g., from the saw chain moving while the engine is idling.

WARNING

Your chain saw is equipped with a chain catcher. It is designed to reduce the risk of personal injury in the event of a thrown or broken saw chain. From time to time, the catcher may be damaged or removed. To reduce the risk of personal injury, do not operate a chain saw with a damaged or missing chain catcher.

WARNING

Inspect antivibration elements periodically. Replace damaged, broken or excessively worn antivibration elements immediately, since they may result in loss of control of the saw. A "sponginess" in the feel of the saw, increased vibration or increased "bottoming" during normal operation may indicate damage, breakage or excessive wear. Antivibration elements should always be replaced in sets. If you have any questions as to whether the antivibration elements should be replaced, consult your STIHL servicing dealer.

If this chain saw is subjected to unusually high loads for which it was not designed (e. g. heavy impact or a fall), always check that it is in good condition before continuing work. Check in particular that the fuel system is tight (no leaks) and that the controls and safety devices are working properly. Do not continue operating this chain saw if it is damaged. In case of doubt, have it checked by your STIHL servicing dealer.

Your chain saw is not designed for prying or shoveling away limbs, roots or other objects. Such use could damage the cutting attachment or AV system.

WARNING

When sawing, make sure that the saw chain does not touch any foreign materials such as rocks, fences, nails and the like. Such objects may be flung off, damage the saw chain or cause the chain saw to kickback.

WARNING

If the rotating saw chain strikes a rock or other hard object, sparks may be created, which can ignite flammabale materials under certain circumstances. Flammable materials can include dry vegetation and brush, particularly when weather conditions are hot and dry. Do not use your chain saw around flammable materials or around dry vegetation or brush when there is a risk of fire or wildfire. Contact your local fire authorities or the U.S. Forestry Service if you have any question about whether vegetation and weather conditions are suitable for the use of a chain saw.

WARNING

Take special care when cutting shattered wood because of the risk of injury from slivers being caught and thrown in your direction.

WARNING

Never modify your muffler. Any modification could cause an increase in heat radiation, sparks or sound level, thereby increasing the risk of fire, burn injury or hearing loss. You may also permanently damage the engine. Have your muffler serviced and repaired by your STIHL servicing dealer only.

WARNING

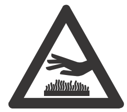
The muffler and other parts of the engine (e.g. fins of the cylinder, spark plug) become hot during operation and remain hot for a while after stopping the engine. To reduce risk of burns, do not touch the muffler and other parts while they are hot. Keep the area around the muffler clean. Remove excess lubricant and all debris such as pine needles, branches or leaves. Let the engine cool down sitting on concrete, metal, bare ground or solid wood (e.g. the trunk of a felled tree) away from any combustible substances.

WARNING

An improperly mounted or damaged cylinder housing or a damaged/deformed muffler shell may interfere with the cooling process of the muffler. To reduce the risk of fire or burn injury, do not continue work with a damaged or improperly mounted cylinder housing or a damaged/deformed muffler shell.

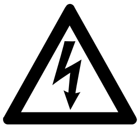
Your muffler is furnished with a spark arresting screen designed to reduce the risk of fire from the emission of hot particles. Never operate your unit with a missing or damaged spark arresting screen. If your gas/oil mix ratio is correct (i.e., not too rich), this screen will normally stay clean as a result of the heat from the muffler and need no service or maintenance. If you experience loss of performance and you suspect a clogged screen, have your muffler maintained by a STIHL servicing dealer. Some state or federal laws or regulations may require a properly maintained spark arrester for certain uses. See the "Maintenance, Repair and Storing" section of these Safety Precautions. Remember that the risk of a brush or forest fire is greater in hot or dry conditions.

WARNING



Some STIHL chain saws are equipped with a catalytic converter, which is designed to reduce the exhaust emissions of the engine by a chemical process in the muffler. Due to this process, the muffler does not cool down as rapidly as conventional mufflers when the engine returns to idle or is shut off. To reduce the risk of fire and burn injuries when using a catalytic converter, always set your chain saw down in the upright position and never locate it where the muffler is near dry brush, grass, wood chips or other combustible materials while it is still hot.

DANGER

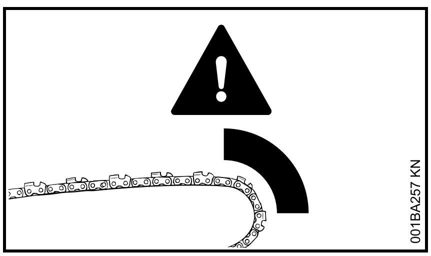


Do not rely on the chain saw 's insulation against electric shock. To reduce the risk of electrocution, never operate this chain saw in the vicinity of any wires or cables (power, etc.) which may be carrying electric current.To reduce risk of electrocution, take extra precautions when cutting near power lines. Have the power switched off before starting cutting work in the immediate vicinity of power lines.

REACTIVE FORCES INCLUDING KICKBACK

WARNING

Reactive forces may occur any time the chain is rotating. Reactive forces can cause serious personal injury.



The powerful force used to cut wood can be reversed and work against the operator. If the rotating saw chain is suddenly and significantly slowed or stopped by contact with any solid object such as a log or branch or is pinched, the reactive forces may occur instantly. These reactive forces may result in loss of control, which, in turn, may cause serious or fatal injury. An understanding of the causes of these reactive forces may help you avoid the element of surprise and loss of control. Surprise contributes to accidents.

The most common reactive forces are:

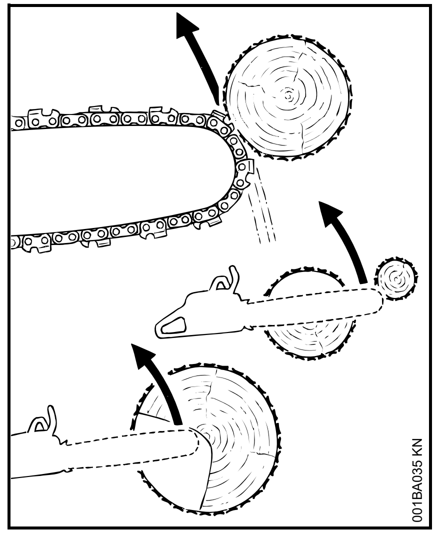
– kickback,
– pushback,
– pull-in.

WARNING

Kickback:

Kickback may occur when the moving saw chain near the upper quadrant of the bar nose contacts a solid object or is pinched.

When this occurs, the energy driving the saw chain can create a force that moves the chain saw in a direction opposite to the saw chain movement at the point where the saw chain is slowed or stopped. This may fling the bar up and back in a lightning fast reaction mainly in the plane of the bar and can cause severe or fatal injury to the operator.



Kickback may occur, for example, when the saw chain near the upper quadrant of the bar nose contacts the wood or is pinched during limbing or when it is incorrectly used to begin a plunge or boring cut.

The greater the force of the kickback reaction, the more difficult it becomes for the operator to control the chain saw. Many factors influence the occurrence and force of the kickback reaction. These include saw chain speed, the speed at which the bar and saw chain contact the object, the angle of contact, the condition of the saw chain and other factors.

The type of bar and saw chain you use is an important factor in the occurrence and force of the kickback reaction. Some STIHL bar and saw chain types are designed to reduce kickback forces. STIHL recommends the use of reduced kickback bars and low kickback chains.

Chain Saw Kickback Standard

The following standard apply with respect to kickback:

– § 5.11 of ANSI/OPEI B175.1-2012

This standard, in the following referred to as "the chain saw kickback standard" sets certain performance and design criteria related to chain saw kickback.

To comply with the chain saw kickback standard:

a) Chain saws with a displacement of less than 3.8 cubic inches (62 cm³)

– must, in their original condition, meet a 45° computer derived kickback angle when equipped with certain cutting attachments,

– and must be equipped with at least two devices to reduce the risk of kickback injury, such as a chain brake, low kickback saw chain, reduced kickback bar, etc.

b) Chain saws with a displacement of 3.8 cubic inches (62 cm³) and above

– must be equipped with at least one device designed to reduce the risk of kickback injury, such as a chain brake, low kickback saw chain, reduced kickback bar, etc.

The computer derived angles for chain saws below 3.8 cubic inches (62 cm³) displacement are measured by applying a computer program to test results from a kickback test machine.

⚠ WARNING

The computer derived angles of the chain saw kickback standard may bear no relationship to actual kickback bar rotation angles that may occur in real life cutting situations.

In addition, features designed to reduce kickback injuries may lose some of their effectiveness when they are no longer in their original condition, especially if they have been improperly maintained. Compliance with the chain saw kickback standard does not automatically mean that in a real life kickback the bar and saw chain will rotate at most 45°.

⚠ WARNING

In order for chain saws below 3.8 cubic inches (62 cm³) displacement to comply with the computed kickback angle requirements of the chain saw kickback standard use only the following cutting attachments:

– bar and saw chain combinations listed as complying in the "Specifications" section of the instruction manual or

– other replacement bar and saw chain combinations marked in accordance with the standard for use on the chain saw or

– replacement saw chain designated "low kickback saw chain."

See the section on "Low Kickback Saw Chain and Reduced Kickback Bars."

Devices for Reducing the Risk of Kickback Injury

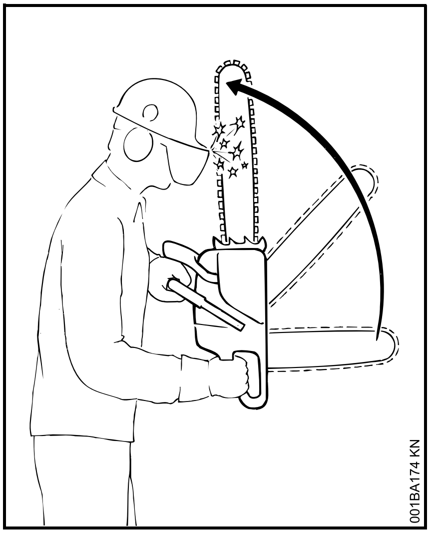
Stihl recommends the use of green labeled reduced kickback bars and low kickback saw chains on your chain saw equipped with a Stihl Quickstop chain brake.

⚠ WARNING

To reduce the risk of injury, never use a chain saw if the chain brake does not function properly. Take the chain saw to your local STIHL servicing dealer. Do not use the chain saw until the problem has been rectified.

STIHL Quickstop Chain Brake

STIHL has developed a saw chain stopping system designed to reduce the risk of injury in certain kickback situations. It is called a Quickstop chain brake.



There are two mechanisms for activating the chain brake if it is in a properly maintained condition:

– manual activation: If a kickback occurs, the chain saw moves upwards towards the user in a rotating motion around the front handle. The brake is designed to engage if the left hand contacts the front guard, which is the activation lever for the brake, and pushes it forward.

– inertia activation: All STIHL chain saws are equipped with an interia Quickstop chain brake. If the kickback impulse is strong enough, this alone is sufficient to engage the brake even without contacting the front hand guard.

WARNING

Never operate your chain saw without a front hand guard. In a kickback situation this guard helps protect your left hand and other parts of your body. In addition, removal of the hand guard on a chain saw equipped with a Quickstop chain brake will disable the activation mechanism of the chain brake.

WARNING

No Quickstop or other chain brake device prevents kickback. These devices are designed to reduce the risk of kickback injury, if activated, in certain kickback situations. In order for the Quickstop to reduce the risk of kickback injury, it must be properly maintained and in good working order. See the chapter of your instruction manual entitled "Chain Brake" and the section "Maintenance, Repair and Storing" at the end of these Safety Precautions. In addition, there must be enough distance between the bar and the operator to ensure that the Quickstop has sufficient time to activate and stop the chain before potential contact with the operator.

WARNING

An improperly maintained chain brake may increase the time needed to stop the saw chain after activation, or may not activate at all.

WARNING

Never run the chain saw above idle speed for more than 3 seconds when the chain brake is engaged or when the saw chain is pinched or otherwise caught in the cut. Clutch slippage can cause excessive heat, leading to severe damage of the motor housing, clutch and oiler component and may interfere with the operation of the chain brake. If clutch slippage in excess of 3 seconds has occurred, allow the motor housing to cool before proceeding and check the operation of your chain brake as described in the chapter entitled "Chain Brake" of your instruction manual. Also make sure that the saw chain is not turning at idle speed (see above at "Important Adjustments").

Low Kickback Saw Chain and Reduced Kickback Bars

STIHL offers a variety of bars and saw chains. STIHL reduced kickback bars and low kickback saw chains are designed to reduce the risk of kickback injury. Other saw chains are designed to achieve higher cutting performance or sharpening ease, but in turn are more prone to kickback.

STIHL has developed a color code system to help you identify the STIHL reduced kickback bars and low kickback saw chains. Cutting attachments with green warning labels on the packaging are designed to reduce the risk of kickback injury. The matching of green marked or labeled chain saws under 3.8 cubic inches (62 cm³) displacement with green labeled bars and green labeled saw chains gives compliance with the computed kickback angle requirements of the chain saw standard when the products are in their original condition. Products with yellow labels are for users with extraordinary cutting needs, having experience and specialized training for dealing with kickback.

STIHL recommends the use of its green labeled reduced kickback bars, green labeled low kickback saw chains and a chain saw equipped with a STIHL Quickstop chain brake for both experienced and inexperienced chain saw users.

Please ask your STIHL dealer to properly match your chain saw with the appropriate bar / saw chain combination to reduce the risk of kickback injury. Green labeled bars and saw chains are recommended for all chain saws.

⚠ WARNING

Use of other, non-listed bar / saw chain combinations may increase kickback forces and the risk of kickback injury. New bar / saw chain combinations may be developed after publication of this literature, which will, in combination with certain chain saws, comply with the chain saw standard as well. Check with your STIHL dealer for such combinations.

⚠ WARNING

Reduced kickback bars and low kickback saw chains do not prevent kickback, but they are designed to reduce the risk of kickback injury. They are available from your STIHL dealer.

⚠ WARNING

Even if your saw is equipped with a Quickstop, a reduced kickback bar and / or low kickback saw chain, this does not eliminate the risk of injury by kickback. Therefore, always observe all safety precautions to avoid kickback situations.

Low Kickback Saw Chain

Some types of saw chains have specially designed components to reduce the force of nose contact kickback. STIHL has developed low kickback saw chain for your chain saw.

A "low kickback saw chain" is a saw chain that has met the kickback performance requirements of ANSI/OPEI B175.1-2012 when tested according to the provisions specified in ANSI/OPEI B175.1-2012.

⚠ WARNING

There are potential chain saw and bar combinations with which low kickback saw chains can be used which have not been specifically certified to comply with the 45° computer derived kickback angle of the chain saw standard. Some low kickback saw chains have not been tested with all chain saw and bar combinations.

⚠ WARNING

A blunt or incorrectly sharpened saw chain may reduce or negate the effects of the design features intended to reduce kickback energy. Improper lowering or sharpening of the depth gauges as well as changing the shape of the cutters may increase the risk and the energy of kickback. Always cut with a properly sharpened saw chain.

Reduced Kickback Bars

STIHL green labeled reduced kickback bars are designed to reduce the risk of kickback injury when used with STIHL green labeled low kickback saw chains.

⚠ WARNING

When used with other, more aggressive saw chains, these bars may be less effective in reducing kickback.

⚠ WARNING

For a properly balanced saw and in order to comply with the chain saw standard, use only bar lengths listed in the specifications chapter of the instruction manual for your chain saw.

To avoid kickback

The best protection from personal injury that may result from kickback is to avoid kickback situations:

1. Hold the chain saw firmly with both hands and maintain a secure grip. Don’t let go.
2. Be aware of the location of the guide bar nose at all times.
3. Never let the nose of the guide bar contact any object. Do not cut limbs with the nose of the guide bar. Be especially careful near wire fences and when cutting small, tough limbs, small size brush and saplings which may easily catch the saw chain.
4. Don't overreach.
5. Don't cut above shoulder height.
6. Begin cutting and continue at full throttle.

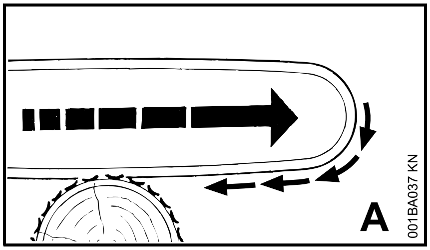
7. Cut only one log at a time.
8. Use extreme caution when reentering a previous cut.
9. Do not attempt to plunge cut if you are not experienced with these cutting techniques.
10. Be alert for shifting of the log or other forces that may cause the cut to close and pinch the saw chain.
11. Maintain saw chain properly. Cut with a correctly sharpened, properly tensioned saw chain at all times.
12. Stand to the side of the cutting path of the chain saw.

Bow Guides

⚠ WARNING

Do not mount a bow guide on any STIHL chain saw. Any chain saw equipped with a bow guide is potentially very dangerous. The risk of kickback is increased with a bow guide because of the increased kickback contact area. Low kickback saw chain will not significantly reduce the risk of kickback injury when used on a bow guide.

A = Pull-in



Pull-in occurs when the saw chain on the bottom of the bar is suddenly stopped when it is pinched, caught or encounters a foreign object in the wood. The reaction of the saw chain pulls the chain saw forward and may cause the operator to lose control.

Pull-in frequently occurs when the bumper spike of the chain saw is not held securely against the tree or limb and when the saw chain is not rotating at full speed before it contacts the wood.

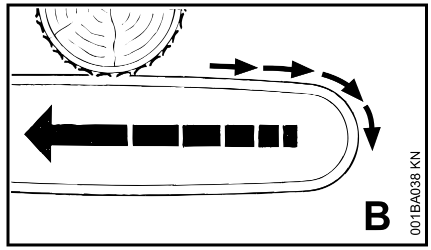
⚠ WARNING

Use extreme caution when cutting small size brush and saplings which may easily catch the saw chain, be whipped towards you or pull you off balance.

To avoid pull-in

1. Always start a cut with the saw chain rotating at full speed and the bumper spike in contact with the wood.
2. The risk of pull-in may also be reduced by using wedges to open the kerf or cut.

B = Pushback



Pushback occurs when the saw chain on the top of the bar is suddenly stopped when it is pinched, caught or encounters a foreign object in the wood. The reaction of the saw chain may drive the chain saw rapidly straight back toward the operator and may cause loss of chain saw control, which, in turn, may cause serious or fatal injury. Pushback frequently occurs when the top of the bar is used for cutting.

To avoid pushback

1. Be alert to forces or situations that may cause material to pinch the top of the saw chain.
2. Do not cut more than one log at a time.
3. Do not twist the chain saw when withdrawing the bar from a plunge cut or underbuck cut because the saw chain can pinch.

Limbing

Limbing is removing the branches from a fallen tree.

WARNING

There is an extreme danger of kickback during the limbing operation. Do not work with the nose of the bar. Be extremely cautious and avoid contacting the log or other limbs with the nose of the guide bar.

Do not stand on a log while limbing it – you may slip or the log may roll.

Start limbing by leaving the lower limbs to support the log off the ground. When underbucking freely hanging limbs, a pinch may result or the limb may fall, causing loss of control. If a pinch occurs, stop the engine and remove the saw by lifting the limb.

WARNING

Be extremely cautious when cutting limbs or logs under tension (spring poles). The limbs or logs could spring back toward the operator and cause loss of control of the saw and severe or fatal injury to the operator.

Bucking

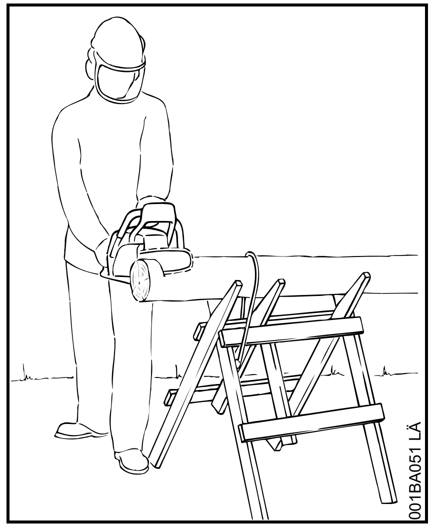
Bucking is cutting a log into sections.

WARNING

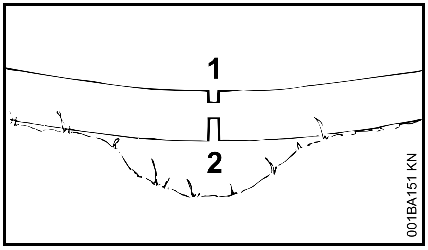
When bucking, do not stand on the log. Make sure the log will not roll downhill. If on a slope, stand on the uphill side of the log. Watch out for rolling logs.

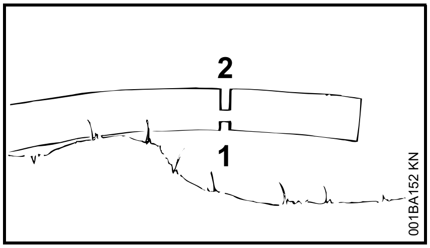
Cut only one log at a time.

Shattered wood should be cut very carefully. Sharp slivers of wood may be caught and flung in the direction of the operator of the saw.



When cutting small logs, place log through "V"-shaped supports on top of a sawhorse. Never permit another person to hold the log. Never hold the log with your leg or foot.





Logs under strain:

Risk of pinching! Always start relieving cut (1) at compression side.Then make bucking cut (2) at tension side. If the saw pinches, stop the engine and remove it from the log.

Only properly trained professionals should work in an area where the logs, limbs and roots are tangled. Working in "blow down" areas is extremely hazardous. Drag the logs into a clear area before cutting. Pull out exposed and cleared logs first.

Felling

Felling is cutting down a tree.

Before felling a tree, consider carefully all conditions which may affect the direction of fall.

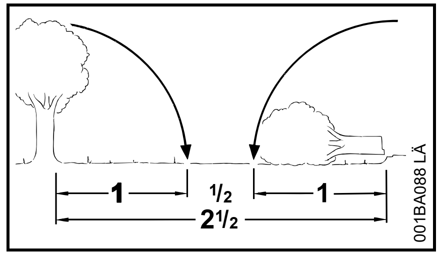
WARNING

There are a number of factors that may affect and change the intended direction of fall, e.g. wind direction and speed, lean of tree, surrounding trees and obstacles, sloping ground, one-sided limb structure, wood structure, decay, snow load, etc. To reduce the risk of severe or fatal injury to yourself or others, look for these conditions prior to beginning the cut, and be alert for a change in direction while the tree is falling.

WARNING

Always observe the general condition of the tree. Inexperienced users should never attempt to cut trees that are decayed or rotted inside or that are leaning or otherwise under tension. There is an increased risk that such trees could snap or split while being cut and cause serious or fatal injury to the operator or bystanders. Also look for broken or dead branches which could vibrate loose and fall on the operator. When felling on a slope, the operator should stand on the uphill side if possible.

Felling Instructions



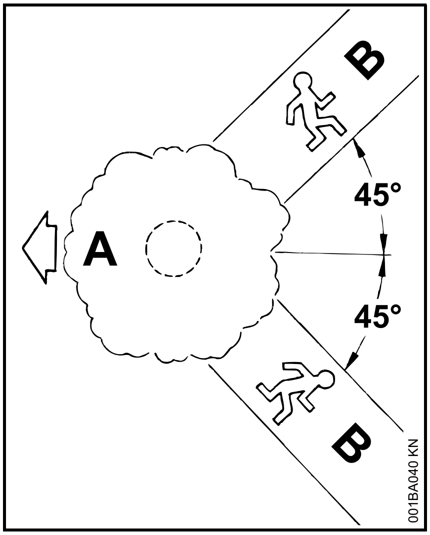
When felling, maintain a distance of at least 2 1/2 tree lengths from the nearest person.

When felling in the vicinity of roads, railways and power lines, etc., take extra precautions. Inform the police, utility company or railway authority before beginning to cut.

WARNING

The noise of your engine may drown any warning call.

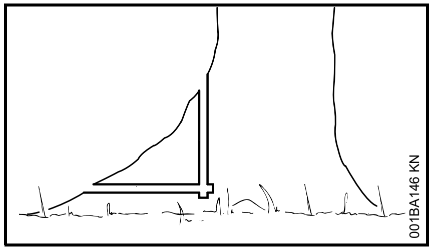
Escape Path



First clear the tree base and work area from interfering limbs and brush and clean its lower portion with an ax.

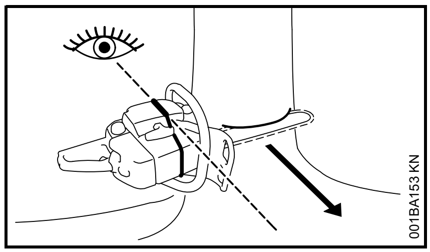
Then, establish two paths of escape (B) and remove all obstacles. These paths should be generally opposite to the planned direction of the fall of the tree (A) and about at a 45° angle. Place all tools and equipment a safe distance away from the tree, but not on the escape paths.

Buttress Roots



If the tree has large buttress roots, cut into the largest buttress vertically first (horizontally next) and remove the resulting piece.

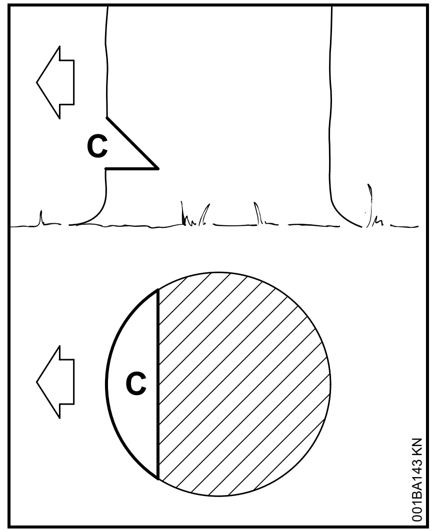
Gunning Sight



When making the felling notch, use the gunning sight on the shroud and housing to check the desired direction of fall:

Position the saw so that the gunning sight points exactly in the direction you want the tree to fall.

Conventional Cut

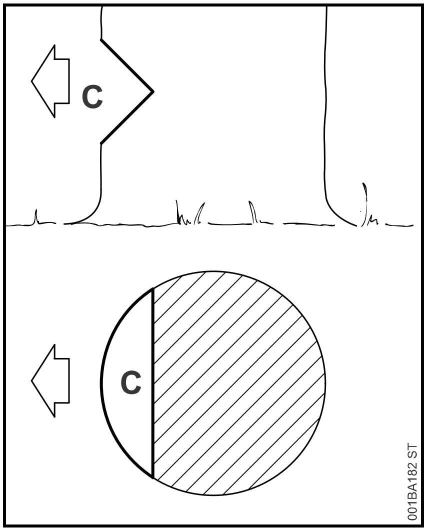


Felling notch (C) – determines the direction of the fall

For a conventional cut:

- Properly place felling notch perpendicular to the line of fall, close to the ground.
- Cut down at approx. 45° angle to a depth of about 1/5 to 1/4 of the trunk diameter.
- Make second cut horizontal.
- Remove resulting 45° piece.

Open-face Technique

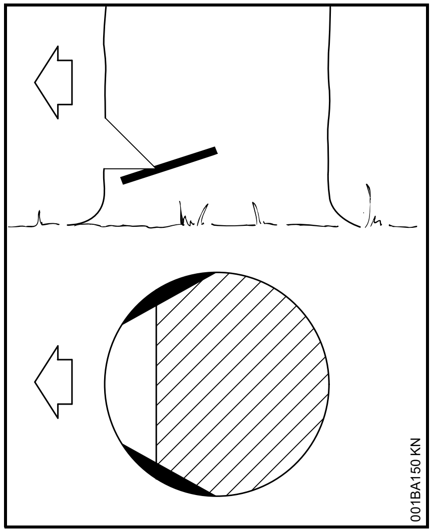


Felling notch (C) – determines the direction of the fall

For an open-face cut:

- Properly place felling notch perpendicular to the line of fall, close to the ground.
- Cut down at approx. 50° angle to a depth of approx.1/5 to 1/4 of the trunk diameter.
- Make second cut from below at approx. 40 degree angle.
- Remove resulting 90° piece.

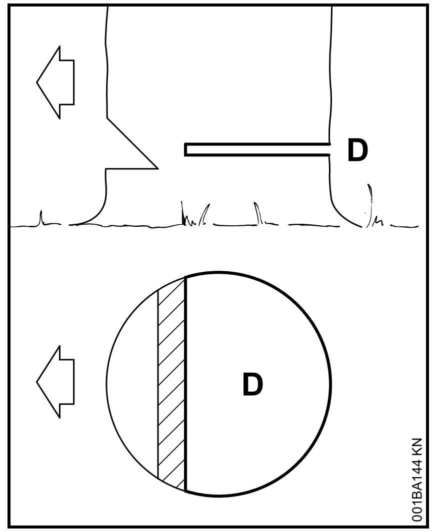
Making Sapwood Cuts



- For medium sized or larger trees make cuts at both sides of the trunk, at same height as subsequent felling cut.
- Cut to no more than width of guide bar.

This is especially important in softwood in summer – it helps prevent sapwood splintering when the tree falls.

D =Felling Cut



Conventional and open-face technique:

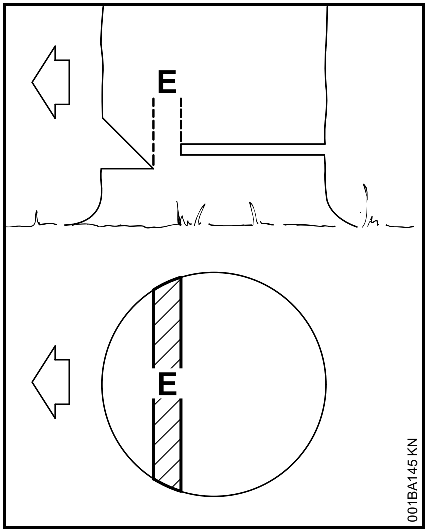
- Begin 1 to 2 inches (2,5 to 5 cm) higher than center of felling notch.
- Cut horizontally towards the felling notch.
- Leave approx.1/10 of diameter uncut. This is the hinge.
- Do not cut through the hinge – you could lose control of the direction of the fall.

Drive wedges into the felling cut where necessary to control the fall.

WARNING

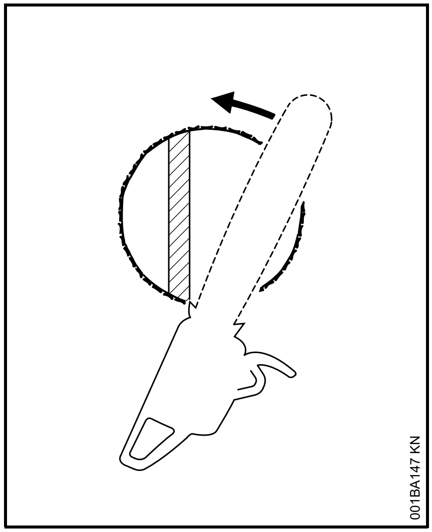
If the tip of the bar contacts a wedge, it may cause kickback. Wedges should be of wood or plastic – never steel, which can damage the chain.

E = Hinge



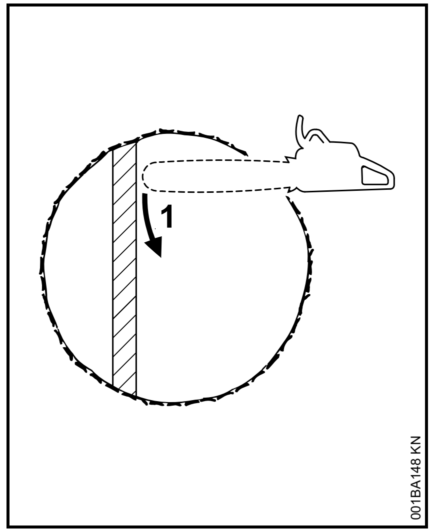
- Helps control the falling tree.
- Do not cut through the hinge – you could lose control of the direction of the fall.

Felling Cut for Small Diameter Trees: Simple Fan Cut



Engage the bumper spikes of the chain saw directly behind the location of the intended hinge and pivot the saw around this point only as far as the hinge. The bumper spike rolls against the trunk.

Felling Cut for Large Diameter Trees

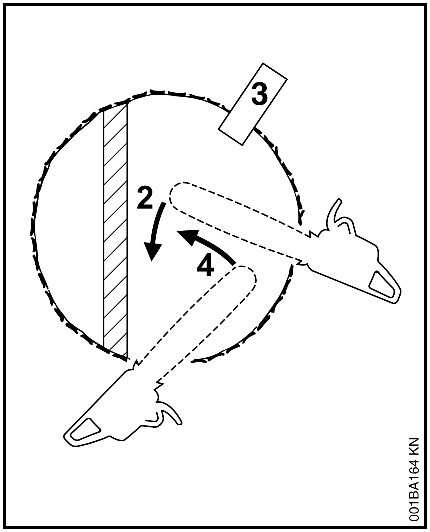


WARNING

Felling a tree that has a diameter greater than the length of the guide bar requires use of either the sectioning felling cut or plunge-cut method. These methods are extremely dangerous because they involve the use of the nose of the guide bar and can result in kickback. Only properly trained professionals should attempt these techniques.

Sectioning Method

For the sectioning method make the first part of the felling cut with the guide bar fanning in toward the hinge. Then, using the bumper spike as a pivot, reposition the saw for the next cut.

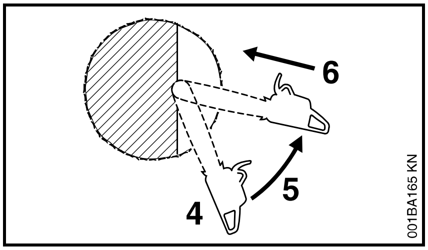


Avoid repositioning the saw more than necessary. When repositioning for the next cut, keep the guide bar fully engaged in the kerf to keep the felling cut straight. If the saw begins to pinch, insert a wedge to open the cut. On the last cut, do not cut the hinge.

Plunge-cut Method

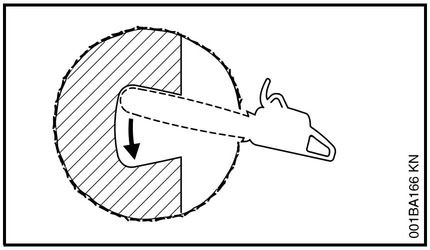
Timber having a diameter more than twice the length of the guide bar requires the use of the plunge-cut method before making the felling cut.

First, cut a large, wide felling notch. Make a plunge cut in the center of the notch.



The plunge cut is made with the guide bar nose. Begin the plunge cut by applying the lower portion of the guide bar nose to the tree at an angle. Cut until the depth of the kerf is about the same as the width of the guide bar. Next, align the saw in the direction in which the recess is to be cut.

With the saw at full throttle, insert the guide bar in the trunk.



Enlarge the plunge cut as shown in the illustration.

WARNING

There is an extreme danger of kickback at this point. Extra caution must be taken to maintain control of the saw. To make the felling cut, follow the sectioning method described previously.

If you are inexperienced with a chain saw, plunge-cutting should not be attempted. Seek the help of a professional.

WARNING

In order to reduce the risk of personal injury, never stand directly behind the tree when it is about to fall, since part of the trunk may split and come back towards the operator (barber-chairing), or the tree may jump backwards off the stump. Always keep to the side of the falling tree. When the tree starts to fall, withdraw the bar, shut off the engine and walk away on the preplanned escape path. Watch out for falling limbs.

WARNING

Be extremely careful with partially fallen trees which are poorly supported. When the tree hangs or for some other reason does not fall completely, set the saw aside and pull the tree down with a cable winch, block and tackle or tractor. If you try to cut it down with your saw, you may be injured.

MAINTENANCE, REPAIR AND STORING

Maintenance, replacement, or repair of the emission control devices and systems may be performed by any nonroad engine repair establishment or individual. However, if you make a warranty claim for a component which has not been serviced or maintained properly, STIHL may deny coverage.

WARNING

Use only identical STIHL replacement parts for maintenance and repair. Use of non-STIHL parts may cause serious or fatal injury.

Strictly follow the maintenance and repair instructions in the appropriate section of your instruction manual. Please refer to the maintenance chart in this manual.

WARNING

Always stop the engine and make sure that the cutting tool is stopped before doing any maintenance or repair work or cleaning the power tool.

WARNING

Do not attempt any maintenance or repair work not described in your instruction manual. Have such work performed by your STIHL servicing dealer only. For example, if improper tools are used to remove the flywheel or if an improper tool is used to hold the flywheel in order to remove the clutch, structural damage to the flywheel could occur and could subsequently cause the flywheel to burst.

Wear gloves when handling or performing maintenance on saw chains.

WARNING

Use the specified spark plug and make sure it and the ignition lead are always clean and in good condition. Always press spark plug boot snugly onto spark plug terminal of the proper size. (Note: If terminal has detachable SAE adapter nut, it must be securely attached.) A loose connection between spark plug terminal and the ignition wire connector in the boot may create arcing that could ignite combustible fumes and cause a fire.

WARNING

Never test the ignition system with the spark plug boot removed from the spark plug or with a removed spark plug, since uncontained sparking may cause a fire.

WARNING

Do not operate your power tool if the muffler is damaged, missing or modified. Never operate your chain saw with missing muffler plugs. An improperly maintained muffler will increase the risk of fire and hearing loss. Your muffler is equipped with a spark-arresting screen to reduce the risk of fire; never operate your power tool if the screen is missing, damaged or clogged. Remember that the risk of a brush or forest fire is greater in hot or dry weather.

In California, it is a violation of § 4442 or § 4443 of the Public Resources Code to use or operate gasoline-powered tools on forest-covered, brush-covered or grass-covered land unless the engine's exhaust system is equipped with a complying spark arrester that is maintained in effective working order. The owner/operator of this product is responsible for properly maintaining the spark arrester. Other states or governmental entities/agencies, such as the U.S. Forest Service, may have similar requirements. Contact your local fire agency or forest service for the laws or regulations relating to fire protection requirements.

Keep the chain, bar and sprocket clean; replace worn sprockets or chains. Keep the chain sharp. You can spot a dull chain when easy-to-cut wood becomes hard to cut and burn marks appear on the wood. Keep the chain at proper tension.

Tighten all nuts, bolts and screws except the carburetor adjustment screws after each use.

WARNING

In order for the chain brake on your STIHL chain saw to properly perform its function of reducing the risk of kickback and other injuries, it must be properly maintained. Like an automobile brake, a chain saw chain brake incurs wear each time it is engaged.

The amount of wear will vary depending upon usage, conditions under which the saw is used and other factors. Excessive wear will reduce the effectiveness of the chain brake and can render it inoperable.

For the proper and effective operation of the chain brake, the brake band and clutch drum must be kept free of dirt, grease and other foreign matter which may reduce friction of the band on the drum.

For these reasons, each STIHL chain saw should be returned to trained personnel such as your STIHL servicing dealer for periodic inspection and servicing of the brake system according to the following schedule:

Heavy usage – every three months,
Moderate usage – twice a year,
Occasional usage – annually.

The chain saw should also be returned immediately for maintenance whenever the brake system cannot be thoroughly cleaned or there is a change in its operating characteristics.

For any maintenance of the emission control system please refer to the maintenance chart and to the limited warranty statement near the end of the instruction manual.

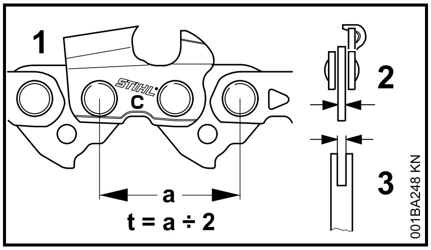
Do not clean your machine with a pressure washer. The solid jet of water may damage parts of the machine.

Store chain saw in a dry place and away from children. Before storing for longer than a few days, always empty the fuel tank (see chapter "Storing the Machine" in the instruction manual).

Cutting Attachment

A cutting attachment consists of the saw chain, guide bar and chain sprocket.

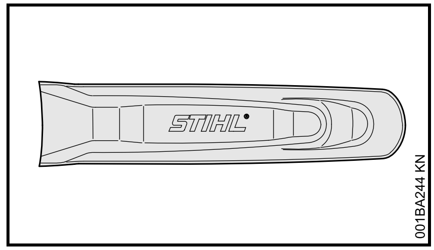
The cutting attachment that comes standard is designed to exactly match the chain saw.



– The pitch (t) of the saw chain (1), chain sprocket and the nose sprocket of the Rollomatic guide bar must match.
– The drive link gauge (2) of the saw chain (1) must match the groove width of the guide bar (3).

If non-matching components are used, the cutting attachment may be damaged beyond repair after a short period of operation.

Chain scabbard



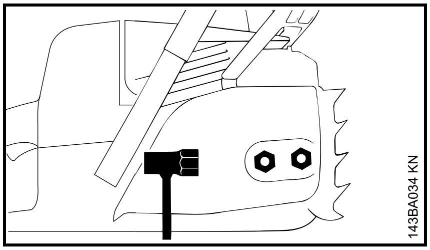
Your saw comes standard with a chain scabbard that matches the cutting attachment.

If you use guide bars of different lengths on the saw, the length of the chain scabbard must be matched to the guide bar to reduce the risk of injury. It should cover the full length of the guide bar.

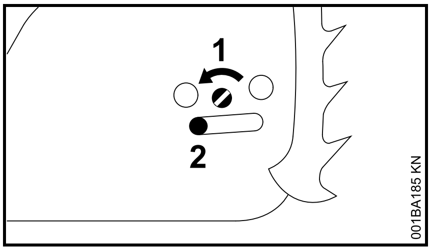
The length of the matching guide bars is marked on the side of the chain scabbard.

Mounting the Bar and Chain

Removing the chain sprocket cover

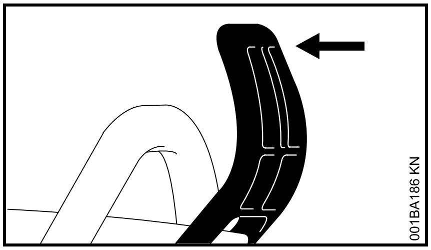


- Unscrew the nuts and take off the chain sprocket cover.



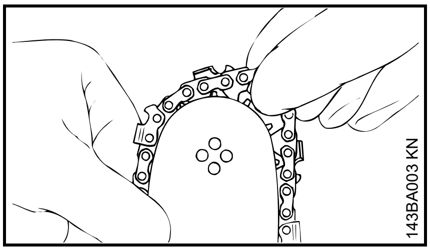
- Turn the screw (1) counterclockwise until the tensioner slide (2) butts against the left end of the housing slot.

Disengaging the chain brake.



- Pull the hand guarad towards the front handle until there is an audible click – the chain brake is disengaged.

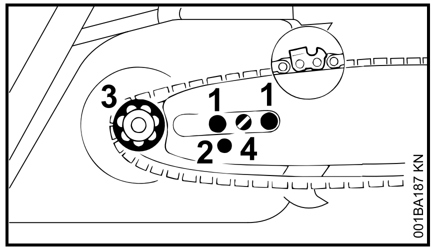
Fitting the chain



⚠ WARNING

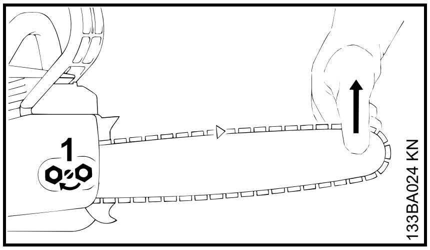
Wear work gloves to protect your hands from the sharp cutters.

- Fit the chain – start at the bar nose.



- Fit the guide bar over the studs (1) – the cutting edges on the top of the bar must point to the right.
- Engage the peg of the tensioner slide in the locating hole (2) –- place the chain over sprocket (3) at the same time.
- Turn the tensioning screw (4)clockwise until there is very little chain sag on the underside of the bar – and the drive link tangs are engaged in the bar groove.
- Refit the sprocket cover and screw on the nuts only fingertight.
- Go to chapter on "Tensioning the Saw Chain"

Tensioning the Chain



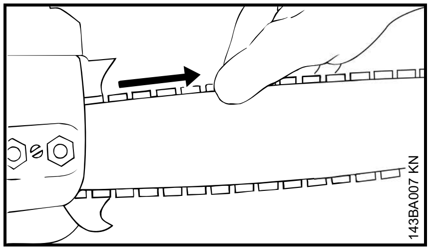
Retensioning during cutting work:

- Shut off the engine.
- Loosen the nuts.
- Hold the bar nose up.
- Use a screwdriver to turn the tensioning screw (1) clockwise until the chain fits snugly against the underside of the bar.
- While still holding the bar nose up, tighten down the nuts firmly.
- Go to "Checking Chain Tension".

A new chain has to be retensioned more often than one that has been in use for some time.

- Check chain tension frequently – see chapter on "Operating Instructions".

Checking Chain Tension



- Shut off the engine.
- Wear work gloves to protect your hands.
- The chain must fit snugly against the underside of the bar and it must still be possible to pull the chain along the bar by hand.
- If necessary, retension the chain.

A new chain has to be retensioned more often than one that has been in use for some time.

- Check chain tension frequently – see chapter on "Operating Instructions".

Fuel

This engine is certified to operate on unleaded gasoline and the STIHL two-stroke engine oil at a mix ratio of 50:1.

Your engine requires a mixture of high-quality gasoline and two-stroke air cooled engine oil.

Use mid-grade unleaded gasoline with a minimum octane rating of 89 ((R+M)/2) and no more than 10% ethanol content.

NOTICE

Fuel with an octane rating below 89 may increase engine temperatures. This, in turn, increases the risk of piston seizure and damage to the engine.

The chemical composition of the fuel is also important. Some fuel additives not only detrimentally affect elastomers (carburetor diaphragms, oil seals, fuel lines, etc.), but magnesium castings and catalytic converters as well. This could cause running problems or damage the engine. For this reason STIHL recommends that you use only quality unleaded gasoline!

NOTICE

Gasoline with an ethanol content of more than 10% can cause running problems and major damage in engines and should not be used.

For further details, see

www.STIHLusa.com/ethanol

The ethanol content in gasoline affects engine speed – it may be necessary to readjust the carburetor if you use fuels with various ethanol contents.

WARNING

To reduce the risk of personal injury from loss of control and/or contact with the running cutting tool, do not use your unit with an incorrect idle adjustment. At correct idle speed, the cutting tool should not move.

If your machine’s idle speed is incorrectly adjusted, have your authorized STIHL servicing dealer check your machine and make the proper adjustments and repairs.

The idle speed and maximum speed of the engine change if you switch from a fuel with a certain ethanol content to a fuel with a much higher or lower ethanol content.

This problem can be avoided by always using fuel with the same ethanol content.

To ensure the maximum performance of your STIHL engine, use a high quality 2-cycle engine oil. To help your engine run cleaner and reduce harmful carbon deposits, STIHL recommends using STIHL HP Ultra 2-cycle engine oil or ask your dealer for an equivalent fully synthetic 2-cycle engine oil.

To meet the requirements of EPA and CARB we recommend to use STIHL HP Ultra oil.

STIHL MotoMix

STIHL recommends the use of STIHL MotoMix. STIHL MotoMix has a high octane rating and ensures that you always use the right gasoline/oil mix ratio.

STIHL MotoMix uses STIHL HP Ultra two-stroke engine oil suited for high performance engines.

For further details, see

www.STIHLusa.com/ethanol

If not using MotoMix, use only STIHL two-stroke engine oil or equivalent high-quality two-stroke engine oils that are designed for use in air cooled two-cycle engines.

The use of non-seasonal gasoline blends may increase the potential for pressure to build in the fuel tank during operation. For example, using a winter blend during the summer will increase pressure in the fuel tank. Always use gasoline blends appropriate to the season, altitude and other environmental factors.

Do not use NMMA or TCW rated (two-stroke water cooled) mix oils or other mix oils that state they are for use in both water cooled and air cooled engines (e.g., outboard motors, snowmobiles, chain saws, mopeds, etc.).

WARNING

Take care when handling gasoline. Avoid direct contact with the skin and avoid inhaling fuel vapor. When filling at the pump, first remove the container from your vehicle and place the container on the ground before filling. To reduce the risk of sparks from static discharge and resulting fire and/or explosion, do not fill fuel containers that are sitting in or on a vehicle or trailer.

The container should be kept tightly closed in order to limit the amount of moisture that gets into the mixture.

The machine's fuel tank should be cleaned as necessary.

Fuel mix ages

If not using MotoMix, only mix sufficient fuel for a few days of work, not to exceed 30 days of storage. Store in approved fuel-containers only. When mixing, pour oil into the container first, and then add gasoline. Close the container and shake it by hand to ensure proper mix of oil and gasoline.

WARNING

Shaking fuel can cause pressure to build in the fuel container. To reduce the risk of fire and severe personal injury or property damage from fuel spraying, allow the fuel container to sit for several minutes before opening. Open the container slowly to release any residual pressures. Never open the fuel container in the vicinity of any ignition source. Read and follow all warnings and instructions that accompany your fuel container.

Gasoline US gal.	Oil (STIHL 50:1 or equivalent high-quality oils) US fl.oz.
1	2.6
2 1/2	6.4
5	12.8

Dispose of empty mixing-oil containers only at authorized disposal locations.

Fueling

WARNING

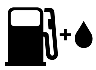
Removing the cap on a pressurized fuel tank can result in gasoline, vapors and fumes being forcefully sprayed out from the tank in all directions. The escaping gasoline, vapors or fumes, sometimes referred to as fuel spraying or "geysering," can cause serious personal injury, including fire and burn injury, or property damage.

Fuel spraying can occur when the engine is hot and the tank is opened while under pressure. It can occur in hot environments even if the engine has not been running. Spraying is more likely to occur when the fuel tank is half full or more.

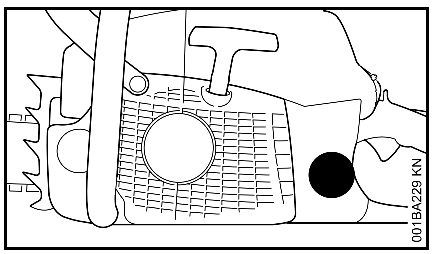
Avoid Injuries from Fuel Spraying.

Always follow the fueling instructions in this manual:

– Treat every fuel tank as if it is pressurized, particularly if it is half full or more.
– Always allow the chain saw to cool adequately before attempting to open the fuel tank or refueling; this will take longer in hot conditions.
– Never remove the cap by turning it directly to the open position. Turn it first approximately 1/8 of a turn counter-clockwise to the vent position to relieve any residual pressure.
– Never open the fuel tank while the engine is still hot or running.
– Never open the fuel tank or re-fuel the saw near any sparks, flames or other ignition sources.
– Pick the right fuel: use only good quality (89 octane or higher), fresh fuel blended for the season.
– Vapor lock: do not remove the fuel cap in an effort to relieve vapor lock. Removing the cap has no effect on vapor lock.
– Be aware that fuel spraying is more likely at higher altitudes.



Preparations



- Before fueling, clean the filler cap and the area around it to ensure that no dirt falls into the tank.
- Position the machine so that the filler cap is facing up.

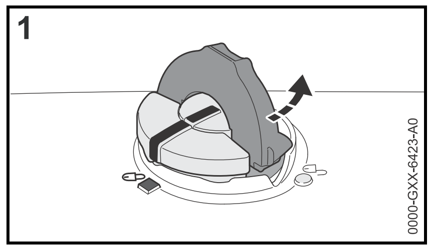
WARNING

In order to reduce the risk of fire and other personal injury from escaping gas vapor and fumes, remove the fuel filler cap slowly and carefully so as to allow any pressure build-up in the tank to release slowly

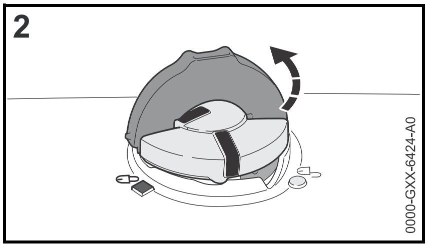
Opening

WARNING

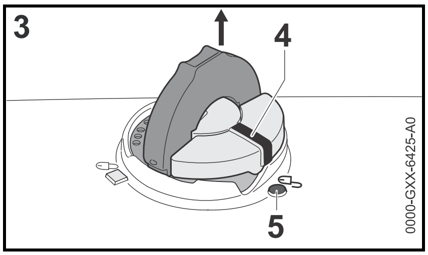
After allowing the chain saw to cool, remove the fuel filler cap slowly and carefully to allow any remaining pressure build-up in the tank to release:



- Flip up the grip and press the cap down firmly (1).



- While maintaining steady, downward pressure, turn the cap slowly counter-clockwise to the vent position (2), approximately a 1/8 turn of the cap.
- If any significant venting occurs, immediately re-seal the tank by turning the cap clockwise to the closed position. Allow the saw to cool further before attempting to open the tank.



- Turn the cap to the open position (3) only after the contents of the tank are no longer under pressure. In the open position, the exterior positioning mark (4) on the cap will line up with the "unlocked" symbol (5) on the fuel tank housing.
- Remove the fuel filler cap.

WARNING

Never remove the cap by turning it directly to the open position. First allow the saw to cool adequately and then release any residual pressure at the vent position (2). Never attempt to remove the cap while the engine is still hot or running.

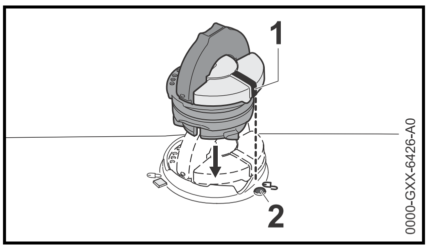
Refueling

Take care not to spill fuel while fueling and do not overfill the tank – leave approximately 1/2" (13 mm) air space.

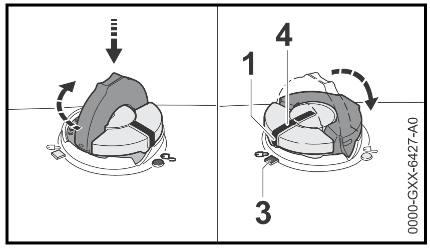
Closing

WARNING

An improperly tightened fuel filler cap can loosen or come off and spill quantities of fuel. To reduce the risk of fuel spillage and fire from an improperly installed fuel cap, correctly position and tighten the cap in the fuel tank opening:



- Raise the grip on the top of the cap until it is upright at a 90° angle. Insert the cap in the fuel tank opening with the exterior positioning mark (1) lined up with the "unlocked" symbol (2) on the fuel tank housing.

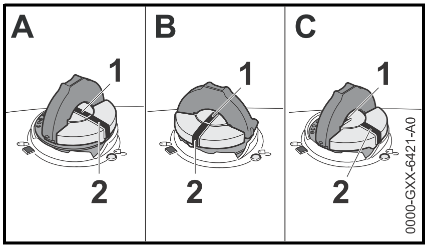


- Using the grip, press the cap down firmly while turning it clockwise to the closed position (approximately 1/4 turn). In the closed position, the interior (4) and exterior (1) positioning marks will align with the "locked" symbol (3) on the fuel tank housing.
- Fold the grip flush with the top of the cap and check for tightness.

WARNING

If the grip does not lie completely flush with the cap or the detent on the grip does not fit in the corresponding recess in the tank opening, or if the cap is loose, the cap is not properly seated and you must repeat the above steps. Also refer to the procedure below for returning the base of the cap to the proper starting position for installation.

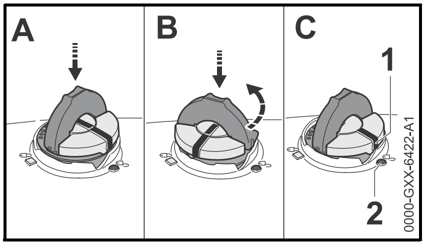
If the filler cap will not engage into the fuel tank housing



If the cap does not drop fully into the fuel tank opening when the positioning marks (1, 2) line up, or if it does not tighten properly when turned, the base of the cap may be prematurely rotated in relation to the top. Such misalignment can result from handling, cleaning or an improper attempt at tightening.

– Illustrations A and B: The base of the cap is prematurely rotated to the closed position and is not in the correct starting position for installation. The tank will not seal in this configuration. Note: in Illustrations A and B, the interior positioning marks (1) are in line with the exterior position marks (2).
– Illustration C: The bottom of the cap is in the correct starting position for installation. Note: In Illustration C, the interior positioning mark (1) is under the grip and not in line with the outer position mark (2).

To return the base of the cap to the proper starting position for installation:



- Drop the cap into the fuel tank opening (A).
- Next, turn the cap counter-clockwise with slight pressure until it drops fully into the fuel tank opening (approximately 1/4 turn) (B). This will rotate the base of the cap into the correct starting position for installation (C). The exterior positioning mark (1) on the cap will line up with the "unlocked" symbol (2) on the fuel tank housing. The interior positioning mark should be under the grip and not in line with the outer positioning mark (1).
- Then, turn the cap clockwise, closing it normally.

If your fuel cap still does not tighten properly, it may be damaged or broken. Stop using the chain saw and take it to your authorized STIHL dealer for repair.

Chain Lubricant

For automatic and reliable lubrication of the chain and guide bar – use only an environmentally compatible quality chain and bar lubricant. Rapidly biodegradable STIHL BioPlus is recommended.

NOTICE

Biological chain oil must be resistant to aging (e.g. STIHL BioPlus), since it will otherwise quickly turn to resin. This results in hard deposits that are difficult to remove, especially in the area of the chain drive and chain. It may even cause the oil pump to seize.

The service life of the chain and guide bar depends on the quality of the lubricant. It is therefore essential to use only a specially formulated chain lubricant.

WARNING

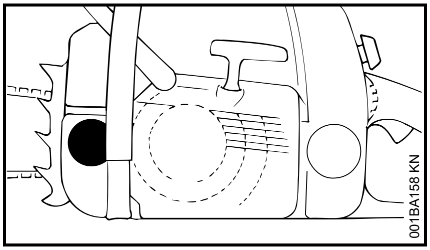
Do not use waste oil. Renewed contact with waste oil can cause skin cancer. Moreover, waste oil is environmentally harmful.

NOTICE

Waste oil does not have the necessary lubricating properties and is unsuitable for chain lubrication.

Filling Chain Oil Tank

Preparations



- Thoroughly clean the oil filler cap and the area around it to ensure that no dirt falls into the tank.
- Position the machine so that the filler cap is facing up.
- Open the filler cap.

Filling chain oil tank

- Refill the chain oil tank every time you refuel.

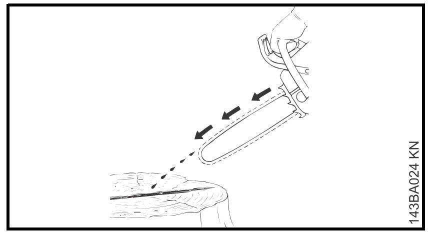
Take care not to spill chain oil while refilling and do not overfill the tank.

- Close the filler cap.

There must still be a small amount of oil in the oil tank when the fuel tank is empty.

If the oil level in the tank does not go down, the reason may be a fault in the oil supply system: Check chain lubrication, clean the oilways, contact your dealer for assistance if necessary STIHL recommends that you have servicing and repair work carried out exclusively by an authorized STIHL servicing dealer.

Checking Chain Lubrication



The saw chain must always spin off a small amount of oil.

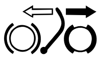
NOTICE

Never operate your machine without chain lubrication. If the saw chain runs dry, the cutting attachment may very quickly be damaged beyond repair. Before starting work, always check the chain lubrication and oil level in the tank.

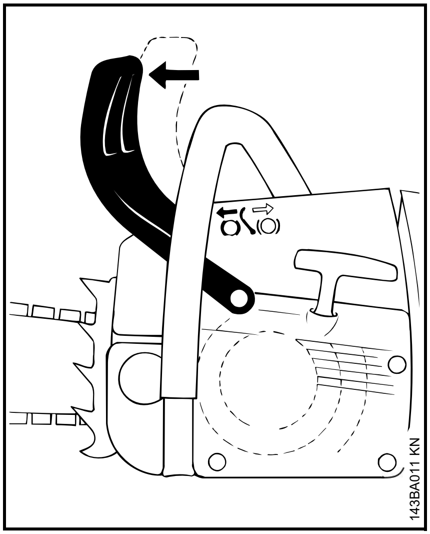
Every new saw chain needs a run-in time of 2 to 3 minutes.

After the saw chain has run in, check the tension of the chain and correct if necessary – see "Checking the chain tension".

Chain Brake



Locking chain with chain brake



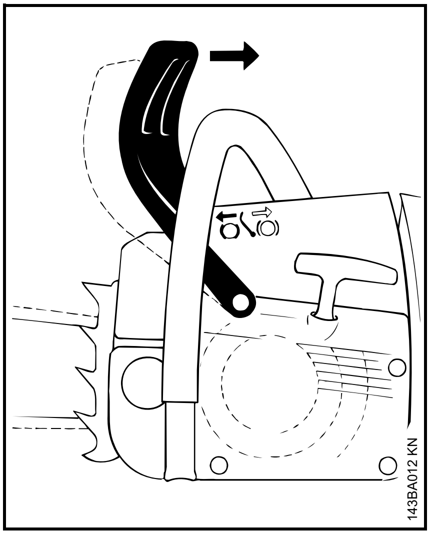
– in an emergency

– when starting

– at idling speed

The chain is stopped and locked when the hand guard is pushed toward the bar nose by the left hand – or when brake is activated by inertia in certain kickback situations.

Releasing the chain brake



- Pull the hand guard back toward the front handle.

NOTICE

Always disengage chain brake before accelerating engine and before starting cutting work. The only exception to this rule is when you check operation of the chain brake.

High revs with the chain brake engaged (chain locked) will quickly damage the powerhead and chain drive (clutch, chain brake).

The chain brake is designed to be activated also by the inertia of the front hand guard

if the forces are sufficiently high. The hand guard is accelerated toward the bar nose - even if your left hand is not behind the hand guard, e.g. during a felling cut. The chain brake will operate only if it has been properly maintained and the hand guard has not been modified in any way.

Check operation of chain brake

Before starting work: Run engine at idle speed, engage the chain brake (push hand guard toward bar nose). Accelerate up to full throttle for no more than 3 seconds – the chain must not rotate. The hand guard must be free from dirt and move freely.

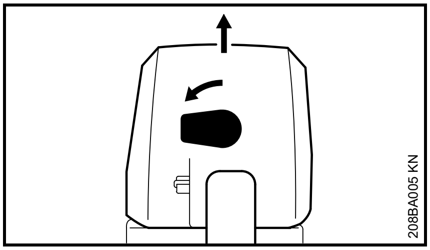
Chain brake maintenance

The chain brake is subject to normal wear. It is necessary to have it serviced and maintained regularly by trained personnel, such as your STIHL servicing dealer, at the following intervals:

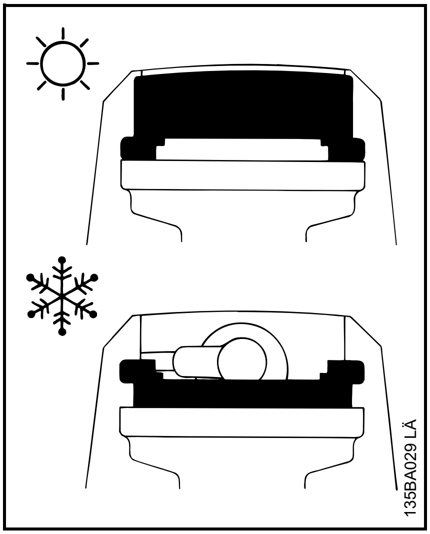
Full-time usage:	every 3 months
Part-time usage:	every 6 months
Occasional usage:	every 12 months

Winter Operation

At temperatures below +50°F (+10°C):



- Squeeze the throttle trigger lockout and move the Master Control lever to |⊷| (cold start).
- Turn the knob above the rear handle 90° counterclockwise.
- Lift off the carburetor box cover vertically.



- Lift out the shutter (in front of spark plug) vertically.
- Turn the shutter 180°.
- Refit the shutter.
- Refit the carburetor box cover and secure it with the knob.

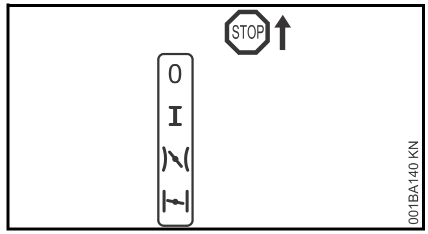
Heated air is now drawn in from around the cylinder to warm the carburetor – this helps prevent carburetor icing.

NOTICE

At temperatures above +70 °F (+ 20°C), always close the shutter again. This is necessary to avoid engine running problems and overheating.

Starting / Stopping the Engine

Positions of the Master Control lever



Stop 0 – engine off – the ignition is switched off

Normal run position I – engine runs or can fire

Starting throttle ( )\( ) – this position is used to start a warm engine. The Master Control lever moves to the normal run position as soon as the throttle trigger is squeezed

Choke shutter closed (|⊷|) – this position is used to start a cold engine

Adjust Master Control lever

To move the Master Control lever from the normal run position (I) to choke closed (|⊷|), press down the throttle lockout and squeeze the throttle trigger at the same time and hold them in that position – now set the Master Control lever.

To select the starting throttle position)\(, move the Master Control lever to choke closed |⊷| first, then push it into the starting throttle position)\(.

The Master Control lever must be in the choke closed position ()⟍() before it can be moved to the starting throttle position (|⌁|).

The Master Control lever moves from the starting throttle position ()⟍() to the normal run position (I) when you press down the throttle lockout and blip the throttle trigger at the same time.

To switch off the engine, set the master control lever to Stop **0**.

Position choke shutter closed |⌁|

– if the engine is cold
– if the engine stalls during opening of throttle after starting
– if the fuel tank was run until empty (engine stopped)

Position starting acceleration)⟍(

– if engine is warm (once the engine has been running for approx. one minute)
– when Engine Begins to Fire
– After ventilation of the combustion chamber, if the engine was flooded

Manual fuel pump

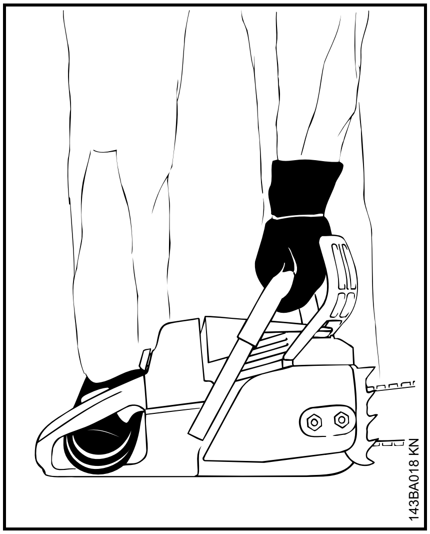
Press the manual fuel pump bulb several times – even if the bulb is still full of fuel:

– When starting for the first time
– if the fuel tank was run until empty (engine stopped)

Holding the Chain Saw

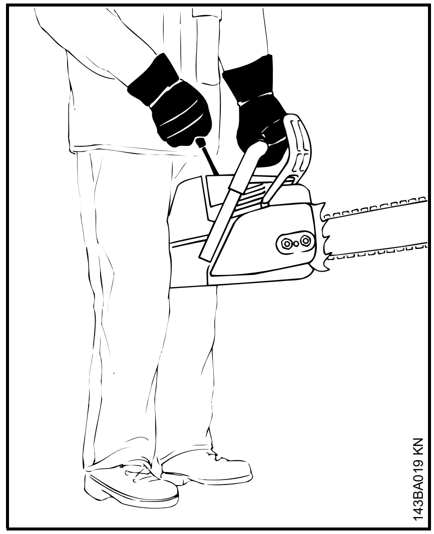
There are two methods of starting the saw.

On the ground



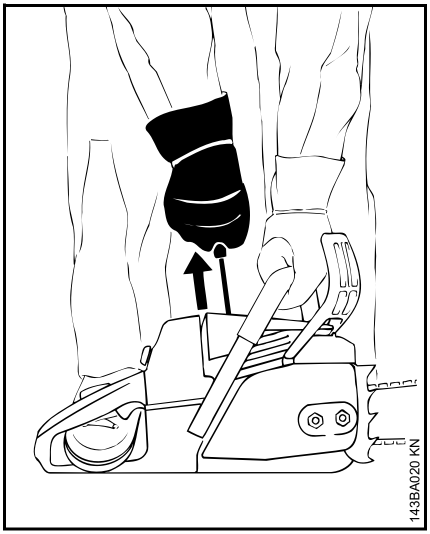
- Place your saw on the ground. Make sure you have a firm footing – check that the chain is not touching any object or the ground.
- Hold the saw firmly on the ground with your left hand on the front handle – your thumb should be under the handle.
- place your right foot through the rear handle.

Between knees



- clamp the rear handle between the knees or thighs
- Hold the front handle firmly with your left hand – your thumb should be under the handle.

Actuating



- Pull the starter grip slowly with your right hand until you feel it engage – and then give it a brisk strong pull and push down the front handle at the same time. Do not pull out the starter rope to full length – **it might otherwise break**. Do not let the starter grip snap back. Guide it slowly back into the housing so that the starter cord can rewind properly.

With a new engine or after a long period of inactivity, it may be necessary to pull the starter rope several times with machines without an additional manual fuel pump to prime the fuel line.

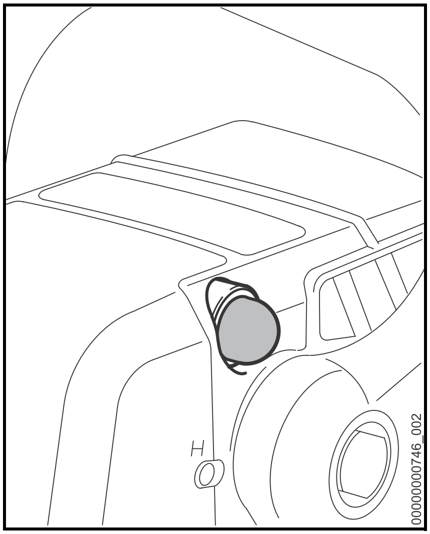
Starting the Chainsaw

WARNING

Bystanders must be well clear of the general work area of the chainsaw.

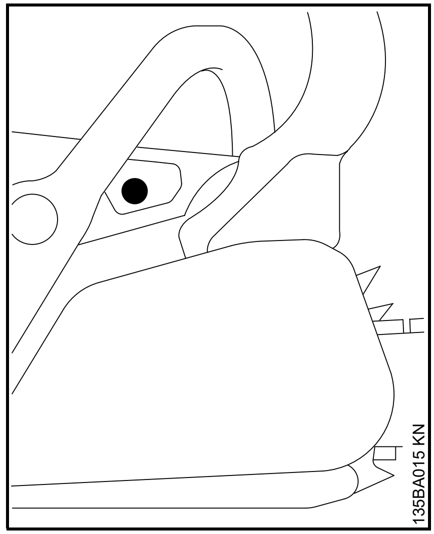
- Observe safety precautions

Versions with manual fuel pump



- Press the manual fuel pump bulb at least five times – even if it is still full of fuel

Versions with decompression valve

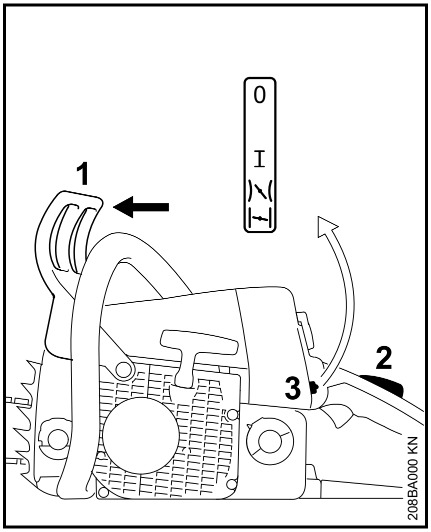


- Press the button, the decompression valve will be opened

It is closed automatically when the engine starts for the first time.

- For this reason, press the button again before each additional starting procedure

For all versions



- Push the hand guard (1) forward – the saw chain is locked.
- Press down throttle trigger lockout (2) and pull the throttle trigger at the same time. Set master control lever (3) to:

Position choke shutter closed |–|

– if engine is cold (even if the engine has stalled during opening of throttle after starting)

Position starting acceleration)\(

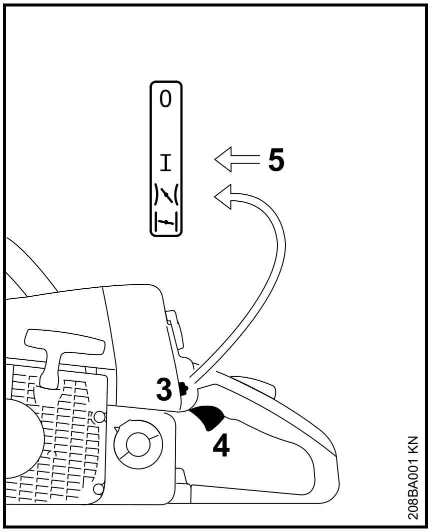
– if engine is warm (once the engine has been running for approx. one minute)

- Hold and start your saw as described.

When Engine Begins to Fire

- Move the master control lever to the starting acceleration position)\(
- Hold and start your chainsaw as described.

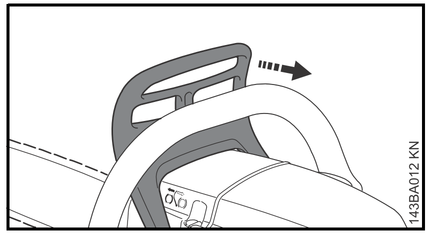
As Soon As the Engine Runs



- Press the throttle trigger lockout and briefly press the throttle trigger (4). The master control lever (3) moves to the run position I and the engine settles down to idling speed.

NOTICE

The engine must be switched to idling speed **immediately** – otherwise, damage to the engine housing and chain brake may occur when the chain brake is locked.



- Pull the hand guard back towards the handlebar.

The chain brake is now disengaged – your saw is ready for operation.

NOTICE

Open the throttle only when the chain brake is off. Increased engine speeds with the chain brake on (saw chain is stationary) will quickly damage the clutch and chain brake.

At Very Low Outside Temperatures

- Let the engine warm up briefly with the throttle slightly open.

Shut off the engine.

- Move the Master Control Lever to the stop position **0**

If Engine Does Not Start

The master control lever was not returned from the position choke shutter closed |–| to starting acceleration)\(in time, the engine may be flooded.

- Move the Master Control Lever to the stop position **0**
- Remove the spark plug – see "Spark Plug".
- Dry the spark plug
- Crank the engine several times with the starter to clear the combustion chamber.
- Refit the spark plug – see "Spark Plug".
- Set the master control lever to starting acceleration – even if the engine is cold
- Restart the engine.

Operating Instructions

During the break-in period

A factory new machine should not be run at high revs (full throttle off load) for the first three tank fillings. This avoids unnecessarily high loads during the break-in period. As all moving parts have to bed in during the break-in period, the frictional resistances in the shortblock are greater during this period. The engine develops its maximum power after about 5 to 15 tank fillings.

During work

NOTICE

Do not make the mixture leaner to achieve an apparent increase in power – this could damage the engine – see "Adjusting the Carburetor".

NOTICE

Open the throttle only when the chain brake is off. Running the engine at high revs with the chain brake engaged (chain locked) will quickly damage the shortblock and chain drive (clutch, chain brake).

Check chain tension frequently

A new saw chain must be retensioned more frequently than one that has been in use already for an extended period.

Chain cold

Tension is correct when the chain fits snugly against the underside of the bar but can still be pulled along the bar by hand. Retension if necessary – see "Tensioning the Saw Chain".

Chain at operating temperature

The chain stretches and begins to sag. The drive links must not come out of the bar groove on the underside of the bar – the chain may otherwise jump off the bar. Retension the chain – see "Tensioning the Saw Chain".

NOTICE

The chain contracts as it cools down. If it is not slackened off, it can damage the crankshaft and bearings.

After a long period of full-throttle operation

After a long period of full-throttle operation, allow engine to run for a while at idle speed so that the heat in the engine can be dissipated by flow of cooling air. This protects engine-mounted components (ignition, carburetor) from thermal overload.

After finishing work

- Slacken off the chain if you have retensioned it at operating temperature during work.

NOTICE

Always slacken off the chain again after finishing work. The chain contracts as it cools down. If it is not slackened off, it can damage the crankshaft and bearings.

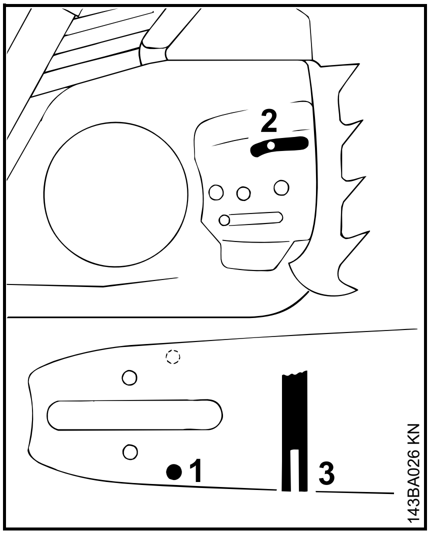
Short-term storage

Wait for engine to cool down. Keep the machine with a full tank of fuel in a dry place, well away from sources of ignition, until you need it again.

Long-term storage

See "Storing the machine"

Taking Care of the Guide Bar



- Turn the guide bar over – every time you sharpen the chain and every time you replace the chain – this helps avoid one-sided wear, especially at the nose and underside of the bar.
- Regularly clean the oil inlet hole (1), the oilway (2) and the bar groove (3).
- Measure the groove depth – with the scale on the filing gauge (special accessory) – in the area used most for cutting.

Chain type	Chain pitch	Minimum groove depth
Picco	1/4" P	0.16" (4.0 mm)
Rapid	1/4"	0.16" (4.0 mm)
Picco	3/8" P	0.20" (5.0 mm)
Rapid	3/8"; 0.325"	0.24" (6.0 mm)
Rapid	0.404"	0.28" (7.0 mm)

If groove depth is less than specified:

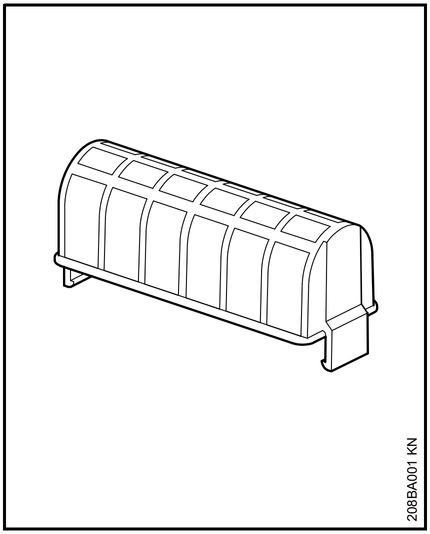
- Replace the guide bar.

The drive link tangs will otherwise scrape along the bottom of the groove – the cutters and tie straps will not ride on the bar rails.

Air Filter System

The air filter system can be adapted to suit different operating conditions by installing a choice of filters. Changing the filter is quick and simple.

The saw comes standard with either a fabric filter or a fleece filter.



Fabric filter

For normal operating conditions and winter operation.

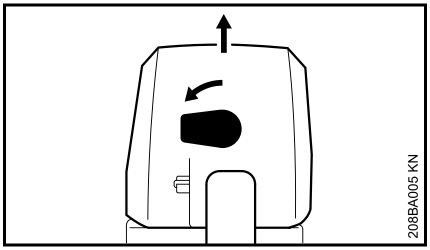
Fleece filter

For dry and very dusty work areas.

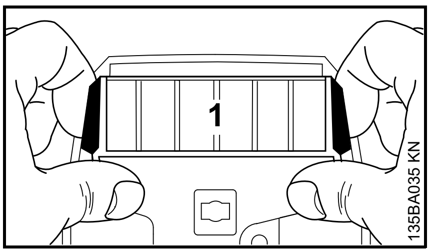
Cleaning the Air Filter

If there is a noticeable loss of engine power

- Press down the throttle trigger lockout and move the Master Control lever to choke closed .



- Turn the knob 90° to the left.
- Lift off the carburetor box cover vertically.
- Always replace damaged filters.
- Clean away loose dirt from around the filter.



- Place fingers behind the air filter (1), press thumbs against the housing and swing the filter in direction of rear handle.

NOTICE

To avoid damaging the filter, do not use tools for removing and installing the air filter.

- Blow out the filter with compressed air from the clean air side.

If the filter fabric is caked with dirt or no compressed air is available:

- Wash the filter in a clean, non-flammable solution (e.g. warm soapy water) and then dry.
- Reinstall the air filter.

Engine Management

Exhaust emissions are controlled by the design of the engine and components (e.g. carburation, ignition, timing and valve or port timing).

Adjusting the Carburetor

General Information

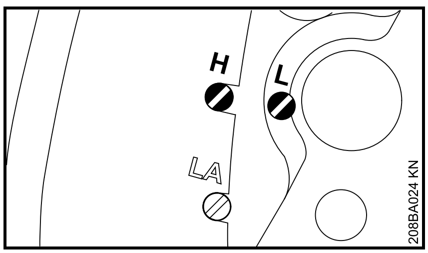
The carburetor comes from the factory with a standard setting.

This setting provides an optimum fuel-air mixture under most operating conditions.

Preparations

- Shut off the engine.
- Check the air filter and clean or replace if necessary.
- Check the spark arresting screen (not in all models, country-specific) in the muffler and clean or replace if necessary.

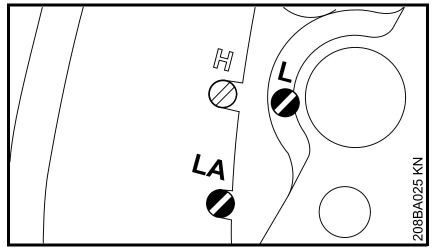
Standard setting



- Turn high speed screw (H) counterclockwise as far as stop (no more than 3/4 turn).
- Turn the low speed screw (L) carefully clockwise as far as stop, then turn it back 1/4 turn.

Adjusting Idle Speed

- Carry out the standard setting.
- Start and warm up the engine.



Engine stops while idling

- Turn the idle speed screw (LA) clockwise until the chain begins to run – then back it off 1/4 turn.

Saw chain runs while engine is idling

- Turn the idle speed screw (LA) counterclockwise until the chain stops moving – then turn it another 1/4 turn in the same direction.

WARNING

If the chain continues moving when the engine is idling, have your saw checked and repaired by your servicing dealer.

Erratic idling behavior, poor acceleration (even though standard setting of low speed screw is correct)

Idle setting is too lean

- Turn the low speed screw (L) counterclockwise, no further than stop, until the engine runs and accelerates smoothly.

It is usually necessary to change the setting of the idle speed screw (LA) after every correction to the low speed screw (L).

Fine Tuning for Operation at High Altitude

A slight correction of the setting may be necessary if engine does not run satisfactorily:

- Carry out the standard setting.
- Warm up the engine.
- Turn high speed screw (H) slightly clockwise (leaner) – no further than stop.

NOTICE

After returning from high altitude, reset the carburetor to the standard setting.

If the setting is too lean there is a risk of engine damage due to insufficient lubrication and overheating.

Spark Plug

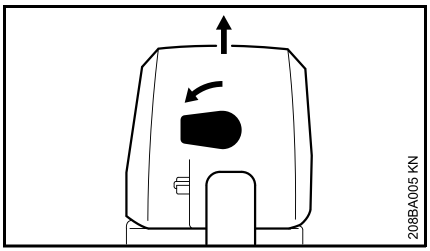
If there is a loss of engine power, the machine is difficult to start or runs poorly at idle, first check the spark plug.

Install a new spark plug after approximately 100 operating hours or earlier if the electrodes are eroded/corroded.

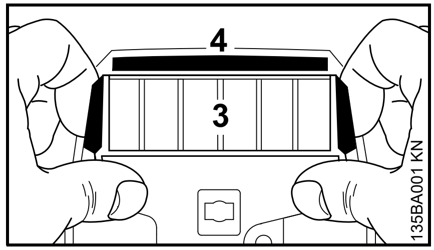
The wrong fuel mix (too much engine oil in the gasoline), a dirty air filter and unfavorable running conditions (mostly at part throttle etc.) affect the condition of the spark plug. These factors cause deposits to form on the insulator nose, which may degrade performance.

Removing the Spark Plug

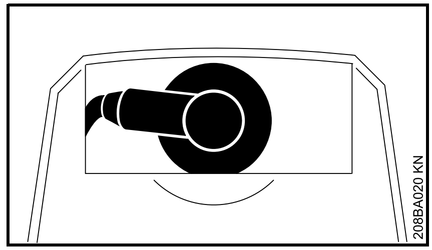
- Depress throttle trigger lockout and throttle trigger at the same time and move the Master Control lever to choke closed ⊢⊣.



- Turn the knob 90° to the left.
- Lift off the carburetor box cover vertically.

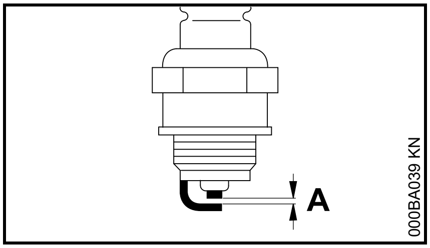


- Place fingers behind the air filter (3), press thumbs against the housing, swing the filter in the direction of the rear handle and remove it.
- Remove the shutter (4).



- Pull off the spark plug boot.
- Unscrew the spark plug.

Checking the spark plug



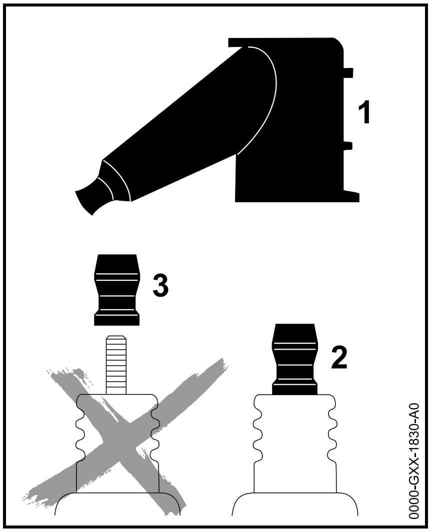
- Clean the spark plug if it is dirty
- Check the electrode gap (A) and readjust if necessary – see "Specifications"
- Use only resistor type spark plugs of the approved range. See the chapter "Specifications" in this instruction manual

Correct the problems that have caused fouling of the spark plug:

– too much oil in fuel mix;

– dirty air filter; or

– unfavorable running conditions, e.g. operating at part throttle.



WARNING

To reduce the risk of fire and burn injury, use only spark plugs authorized by STIHL. Always press the spark plug boot (1) firmly and securely onto the spark plug terminal (2).

Do not use a spark plug with a detachable SAE adapter terminal (3). Arcing may occur that could ignite combustible fumes and cause a fire. This can result in serious injuries or damage to property.

- Only use resistor type spark plugs with solid, non-threaded terminals

Installing the spark plug

- Screw home the spark plug, fit the boot and press it down firmly.
- Install the shutter and air filter.
- Fit the carburetor box cover.

Engine Running Behavior

If engine running behavior is unsatisfactory even though the air filter is clean and the carburetor is properly adjusted, the cause may be the muffler.

Have the muffler checked for contamination (carbonization) by your servicing dealer.

STIHL recommends that you have servicing and repair work carried out exclusively by an authorized STIHL servicing dealer.

Storing the Machine

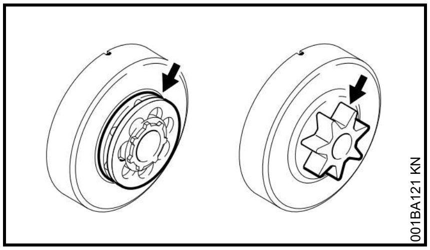
For periods of about 30 days or longer

- Drain and clean the fuel tank in a well-ventilated area.
- Dispose of fuel properly in accordance with local environmental requirements.
- If a manual fuel pump is fitted: Press the manual fuel pump at least 5 times.
- Start the engine and run it at idling speed until it stops.
- Remove saw chain and guide bar; clean and spray with protective oil
- Thoroughly clean the machine - pay special attention to the cylinder fins and air filter
- When using biological chain oil (e.g. STIHL BioPlus), fill the lubricant oil tank
- Store the machine in a dry and secure location Keep out of the reach of children and other unauthorized persons

Checking and Replacing the Chain Sprocket

- Remove chain sprocket cover, saw chain and guide bar.
- Release chain brake – pull hand guard against the front handle

Fit new chain sprocket

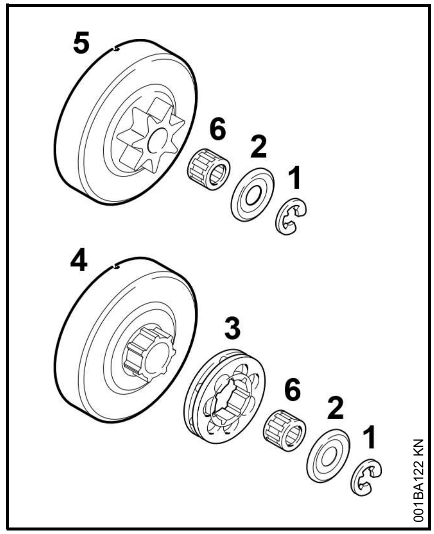


– after use of two saw chains or earlier

– if the wear marks (arrows) are deeper than 0.02 in (0.5 mm) – otherwise the service life of the saw chain is reduced – use check gauge (special accessory) to test

Using two saw chains in alternation helps preserve the chain sprocket.

Use only STIHL original chain sprockets to help ensure reliable functioning of the chain brake.



- Use a screwdriver to remove the E-clip (1)
- Remove the washer (2)
- Remove rim sprocket (3) (if so equipped), clutch drum (4) and needle cage (6).
- Inspect transport profile for rim sprocket on the clutch drum (4) – if there are also heavy signs of wear, also replace the clutch drum
- Remove clutch drum with integrated spur chain sprocket (5) (if so equipped) including needle cage (6) from the crankshaft – for powerheads with Quickstop Plus chain brake system, depress throttle trigger lockout beforehand

Install spur chain sprocket / rim sprocket

- Clean crankshaft stub and needle cage and lubricate with STIHL lubricant (special accessory)
- Slide needle cage onto the crankshaft stub
- After refitting, turn the clutch drum and/or spur chain sprocket approx. 1 full turn so that the carrier for the oil pump drive engages
- Refit the rim sprocket – cavities toward the outside
- Refit washer and E-clip on the crankshaft

Maintaining and Sharpening the Saw Chain

Sawing effortlessly with a properly sharpened saw chain

A properly sharpened saw chain cuts through wood effortlessly even with very little pushing.

Never use a dull or damaged saw chain – this leads to increased physical strain, increased vibration load, unsatisfactory cutting results and increased wear.

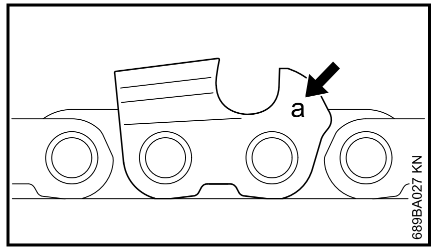
- Clean the saw chain
- Check the saw chain for cracks and damaged rivets
- Replace damaged or worn chain components and adapt these parts to the remaining parts in terms of shape and level of wear – rework accordingly

Carbide-tipped (Duro) saw chains are especially wear-resistant. For an optimal sharpening result, STIHL recommends STIHL servicing dealers.

WARNING

Compliance with the angles and dimensions listed below is absolutely necessary. An improperly sharpened saw chain – especially depth gauges that are too low – can lead to increased kickback tendency of the chain saw – **risk of injury**!

Chain pitch



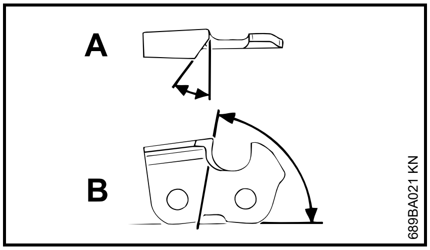
The chain pitch marking (**a**) is embossed in the area of the depth gauge of each cutter.

Marking (**a**)	Chain pitch	
	Inches	mm
7	1/4 P	6.35
1 or 1/4	1/4	6.35
6, P or PM	3/8 P	9.32
2 or 325	0.325	8.25
3 or 3/8	3/8	9.32
4 or 404	0.404	10.26

The diameter of file to be used depends on the chain pitch – see table "Sharpening tools".

The angles of the cutter must be maintained during resharpening.

Sharpening and side plate angles



A Sharpening angle

STIHL saw chains are sharpened with a 30° sharpening angle. Ripping chains, which are sharpened with a 10° sharpening angle, are exceptions. Ripping chains have an X in the designation.

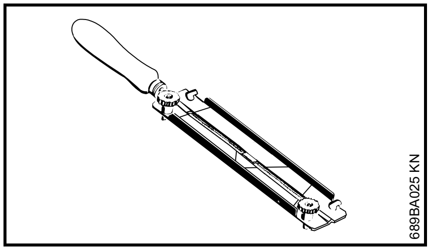
B Side plate angle

The correct side plate angle results automatically when the specified file holder and file diameter are used.

Tooth shapes	Angle (°) A	B
Micro = semi-chisel tooth, e. g., 63 PM3, 26 RM3, 36 RM	30	75
Super = full chisel tooth, e. g., 63 PS3, 26 RS, 36 RS3	30	60
Ripping chain, e. g., 63 PMX, 36 RMX	10	75

The angles must be identical for all cutters in the saw chain. Varying angles: Rough, uneven running of the saw chain, increased wear – even to the point of saw chain breakage.

File holder

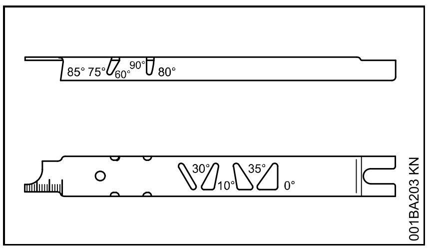


- **Use a file holder**

Always use a file holder (special accessory, see table "Sharpening tools") when sharpening saw chains by hand. File holders have markings for the sharpening angle.

Use only special saw chain files! Other files are unsuitable in terms of shape and type of cutting.

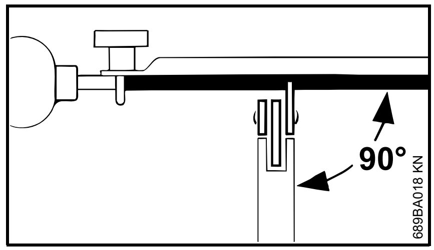
To check the angles

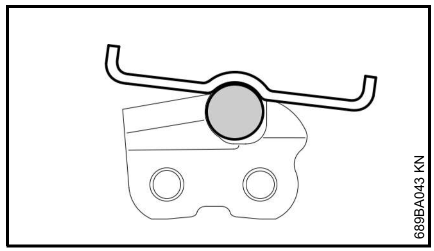


STIHL filing gauge (special accessory, see table "Sharpening tools") – a universal tool for checking sharpening and side plate angles, depth gauge setting, and tooth length, as well as cleaning grooves and oil inlet holes.

Proper sharpening

- Select sharpening tools in accordance with chain pitch
- Clamp guide bar if necessary
- Block saw chain – push the hand guard forward
- To advance the saw chain, pull the hand guard toward the handlebar: The chain brake is disengaged. With the Quickstop Plus chain brake system, additionally press the throttle trigger lockout
- Sharpen frequently, removing little material – two or three strokes of the file are usually sufficient for simple resharpening





- Guide the file: **horizontally** (at a right angle to the side surface of the guide bar) in accordance with the specified angle – according to the markings on the file holder – rest the file holder on the tooth head and the depth gauge

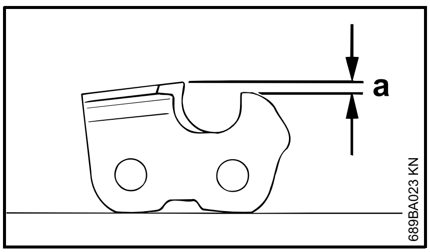
- File only from the inside outward
- The file only sharpens on the forward stroke – lift the file on the backstroke
- Do not file tie straps and drive links
- Rotate the file a little periodically in order to avoid uneven wear
- To remove file burr, use a piece of hardwood
- Check angle with file gauge

All cutters must be equally long.

With varying cutter lengths, the cutter heights also vary and cause rough running of the saw chain and chain breakage.

- All cutters must be filed down equal to the length of the shortest cutter – ideally, one should have this done by a servicing dealer using an electric sharpener

Depth gauge setting



The depth gauge determines the depth to which the cutter penetrates the wood and thus the chip thickness.

a Required distance between depth gauge and cutting edge

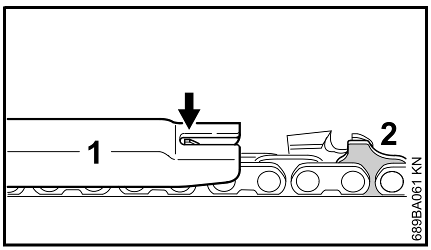
When cutting softwood outside of the frost season, the distance can be increased by up to 0.2 mm (0.008").

Chain pitch		Depth gauge Distance (a)	
Inches	(mm)	mm	(Inches)
1/4 P	(6.35)	0.45	(0.018)
1/4	(6.35)	0.65	(0.026)
3/8 P	(9.32)	0.65	(0.026)
0.325	(8.25)	0.65	(0.026)
3/8	(9.32)	0.65	(0.026)
0.404	(10.26)	0.80	(0.031)

Lowering the depth gauges

The depth gauge setting is lowered when the cutter is sharpened.

- Check the depth gauge setting after each sharpening

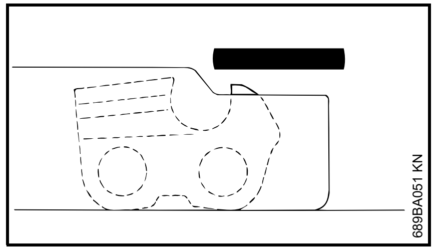


- Lay the appropriate file gauge (1) for the chain pitch on the saw chain and press it against the cutter to be checked – if the depth gauge protrudes past the file gauge, the depth gauge must be reworked

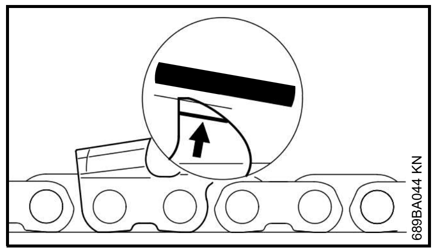
Saw chains with humped drive link (2) – upper part of the humped drive link (2) (with service mark) is lowered at the same time as the depth gauge of the cutter.

WARNING

The rest of the humped drive link must not be filed; otherwise, this could increase the tendency of the chain saw to kick back.



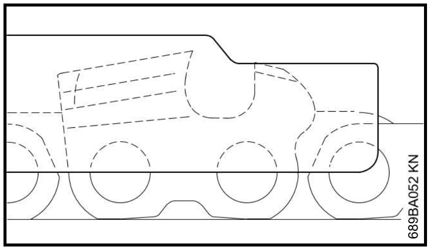
- Rework the depth gauge so that it is flush with the file gauge



- Afterwards, dress the leading edge of the depth gauge parallel to the service mark (see arrow) – when doing this, be careful not to further lower the highest point of the depth gauge

WARNING

Depth gauges that are too low increase the kickback tendency of the chain saw.



- Lay the file gauge on the saw chain – the highest point of the depth gauge must be flush with the file gauge

- After sharpening, clean the saw chain thoroughly, removing any filings or grinding dust – lubricate the saw chain thoroughly
- In the event of extended periods of disuse, store saw chains in cleaned and oiled condition

Sharpening tools (special accessories)

Chain pitch		Round file Ø		Round file	File holder	File gauge	Taper square file	Sharpening set 1)
Inches	(mm)	mm	(Inches)	Part number	Part number	Part number	Part number	Part number
1/4 P	(6.35)	3.2	(1/8)	5605 771 3206	5605 750 4300	0000 893 4005	0814 252 3356	5605 007 1000
1/4	(6.35)	4.0	(5/32)	5605 772 4006	5605 750 4327	1110 893 4000	0814 252 3356	5605 007 1027
3/8 P	(9.32)	4.0	(5/32)	5605 772 4006	5605 750 4327	1110 893 4000	0814 252 3356	5605 007 1027
0.325	(8.25)	4.8	(3/16)	5605 772 4806	5605 750 4328	1110 893 4000	0814 252 3356	5605 007 1028
3/8	(9.32)	5.2	(13/64)	5605 772 5206	5605 750 4329	1110 893 4000	0814 252 3356	5605 007 1029
0.404	(10.26)	5.5	(7/32)	5605 772 5506	5605 750 4330	1106 893 4000	0814 252 3356	5605 007 1030

1) consisting of file holder with round file, taper square file and file gauge

Maintenance and Care

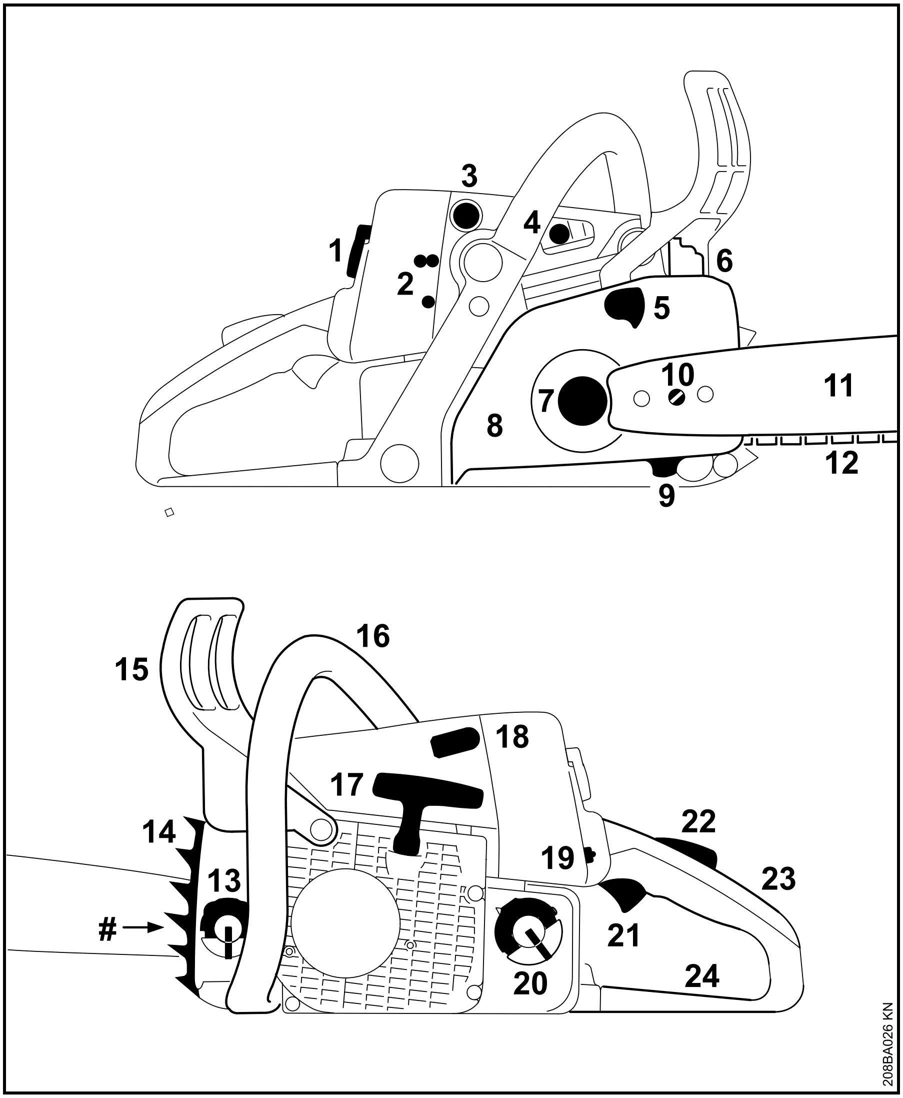
The following information applies under normal operating conditions. The specified intervals must be shortened accordingly when working for longer than normal each day or under difficult conditions (extensive dust, highly resinous lumber, lumber from tropical trees, etc.). If the machine is only used occasionally, the intervals can be extended accordingly.		Before starting work	At the end of work and/or daily	Whenever tank is refilled	Weekly	Monthly	Yearly	If faulty	If damaged	As required
Complete machine	Visual inspection (condition, leaks)	X		X						
	Clean		X							
Throttle trigger, throttle trigger lockout, choke lever, choke control, stop switch, master control lever (dependent on equipment)	Function tests	X		X						
Chain brake	Function tests	X		X						
	Have checked by a specialist dealer[1)]									X
Manual fuel pump (if present)	Check	X								
	Have repaired by a specialist dealer[1)]								X	
Fuel pick-up body / filter in fuel tank	Check					X				
	Clean, replace filter insert					X		X		
	Replace						X		X	X
Fuel tank	Clean					X				
Lubricating oil tank	Clean					X				
Chain lubrication	Check	X								
Saw chain	Check, pay attention to sharpness	X		X						
	Check chain tension	X		X						
	Sharpen									X
Guide bar	Check (wear, damage)	X								
	Clean and turn over									X
	Deburr				X					
	Replace								X	X
Chain sprocket	Check				X					
Air filter	Clean							X		X
	Replace after every 25 hours of operation									

The following information applies under normal operating conditions. The specified intervals must be shortened accordingly when working for longer than normal each day or under difficult conditions (extensive dust, highly resinous lumber, lumber from tropical trees, etc.). If the machine is only used occasionally, the intervals can be extended accordingly.		Before starting work	At the end of work and/or daily	Whenever tank is refilled	Weekly	Monthly	Yearly	If faulty	If damaged	As required
Anti-vibration elements	Check	X						X		
	Have them replaced by a servicing dealer[1]								X	
Air intake on fan housing	Clean		X		X					X
Cylinder fins	Clean		X			X				
Carburetor	Check idle adjustment – saw chain must not rotate	X		X						
	Set the idle speed, have the chain saw repaired by a servicing dealer[1]									X
Spark plug	Adjust electrode gap							X		
	Replace after 100 hours of operation									
All accessible screws, nuts and bolts (not adjusting screws)	Tighten[2]									X
Spark arresting screen in muffler	Check presence	X								
	Check and Clean[1]						X			
Chain catcher	Check	X								
	Replace								X	
Safety information label	Replace								X	

1) STIHL recommends STIHL servicing dealers

2) During initial use of professional chain saws (with a power output of 3.4 kW or more), tighten the cylinder block screws after 10 to 20 hours of operation

Main Parts



1 Carburetor Box Cover Twist Lock
2 Carburetor Adjusting Screws
3 Manual Fuel Pump[1)]
4 Decompression Valve (Automatically Resetting)[1)]
5 Chain Brake
6 Muffler
7 Chain Sprocket
8 Chain Sprocket Cover
9 Chain Catcher
10 Chain Tensioner
11 Guide Bar
12 Oilomatic Saw Chain
13 Oil Filler Cap
14 Bumper Spike
15 Front Hand Guard
16 Front Handle (Handlebar)
17 Starter Grip
18 Spark Plug Boot
19 Master Control Lever
20 Fuel Filler Cap
21 Throttle Trigger
22 Throttle Trigger Lockout
23 Rear Handle
24 Rear Hand Guard
\# Serial Number

543485265

1) Depending on Model

Definitions

1 **Carburetor Box Cover Twist Lock**
Lock for carburetor box cover.

2 **Carburetor Adjusting Screws**
For tuning the carburetor.

3 **Manual Fuel Pump**
Provides additional fuel feed for a cold start.

4 **Decompression Valve (Automatically Resetting)**
Releases compression pressure to make engine starting easier – when activated.

5 **Chain Brake**
A device to stop the rotation of the chain. Is activated in a kickback situation by the operator's hand or by inertia.

6 **Muffler**
Reduces engine exhaust noises and diverts exhaust gases away from operator.

7 **Chain Sprocket**
The toothed wheel that drives the saw chain.

8 **Chain Sprocket Cover**
Covers the clutch and chain sprocket.

9 **Chain Catcher**
Helps to reduce the risk of operator contact by a chain when it breaks or comes off the bar.

10 **Chain Tensioner**
Permits precise adjustment of chain tension.

11 **Guide Bar**
Supports and guides the saw chain.

12 **Oilomatic Saw Chain**
A loop consisting of cutters, tie straps and drive links.

13 **Oil Filler Cap**
For closing the oil tank.

14 **Bumper Spike**
Toothed stop for holding saw steady against wood.

15 **Front Hand Guard**
Provides protection against projecting branches and helps prevent left hand from touching the chain if it slips off the handlebar. It also serves as the lever for chain brake activation.

16 **Front Handle (Handlebar)**
Handlebar for the left hand at the front of the saw.

17 **Starter Grip**
The grip of the pull starter, for starting the engine.

18 **Spark Plug Boot**
Connects the spark plug with the ignition lead.

19 **Master Control Lever**
Lever for starting, operating and stop-engine positions.

20 **Fuel Filler Cap**
For closing the fuel tank.

21 **Throttle Trigger**
Controls the speed of the engine.

22 **Throttle Trigger Lockout**
Must be depressed before the throttle trigger can be activated.

23 **Rear Handle**
The support handle for the right hand, located at the rear of the saw.

24 **Rear Hand Guard**
Gives added protection to operator's right hand.

Guide Bar Nose
The exposed end of the guide bar. (not illustrated, see chapter "Tensioning the Saw Chain")

Clutch
Couples engine to chain sprocket when engine is accelerated beyond idle speed. (not illustrated)

Anti-Vibration System
The anti-vibration system includes a number of anti-vibration elements designed to reduce the transmission of vibrations created by the engine and cutting attachment to the operator's hands. (not illustrated)

Specifications

EPA / CEPA

The Emission Compliance Period referred to on the Emissions Compliance Label indicates the number of operating hours for which the engine has been shown to meet Federal emission requirements.

Category

A = 300 hours

B = 125 hours

C = 50 hours

Engine

STIHL single cylinder two-stroke engine

MS 250

Displacement:	2.77 cu. in 45.4 cc
Bore:	1.7 in (42.5 mm)
Stroke:	1.26 in (32 mm)
Engine power to ISO 7293:	3.0 hp (2.2 kW) at 10,000 rpm
Idle speed:	2,800 rpm
Max. permissible speed (with cutting attachment):	14,000 rpm

Ignition System

Electronic magneto ignition

Spark plug (resistor type):	Bosch WSR 6 F
Electrode gap:	0.02 in (0.5 mm)

Fuel System

All position diaphragm carburetor with integral fuel pump

Fuel tank capacity: 15.9 fl.oz (0.47 l)

Chain Lubrication

Fully automatic, speed-controlled oil pump with rotary piston

Oil tank capacity: 6.8 fl.oz (0.2 l)

Weight

dry, without bar and chain

MS 250: 10.1 lbs (4.6 kg)

Cutting Attachments

Recommended cutting attachments conforming with the 45 degree computed kickback angle requirement of Section 5.11 of ANSI/OPEI B175.1-2012 when used on this model chain saw (see the chapter on "Safety Precautions and Working Techniques"):

Rollomatic E guide bars

Reduced kickback STIHL guide bars (with green label)

Bar lengths:	40, 45 cm (16, 18 in.)
Pitch:	.325" (8.25 mm)
Groove width:	1.6 mm (0.063 in.)
Sprocket nose:	11-tooth

Actual cutting length will be less than listed bar length.

Saw chain .325"

Low kickback STIHL saw chain (with green label)

Rapid Micro 3 (26 RM3) Type 3634

Pitch:	.325" (8.25 mm)
Drive link gauge:	1.6 mm (0.063 in)

Chain sprocket

7-tooth for .325" (spur sprocket)

To comply with the 45 degree computed kickback angle requirement of Section 5.11 of ANSI/OPEI B175.1-2012, use replacement saw chains listed above or elsewhere by STIHL as conforming with that requirement when used on this model chain saw or use saw chains classified as "low kickback" in accordance with ANSI/OPEI B175.1-2012.

Since new bar/chain combinations may be developed after publication of this Manual, ask your STIHL dealer for the latest STIHL recommendations.

Ordering Spare Parts

Please enter your saw model, serial number as well as the part numbers of the guide bar and saw chain in the spaces provided. This will make re-ordering simpler.

The guide bar and saw chain are subject to normal wear and tear. When purchasing these parts, always quote the saw model, the part numbers and names of the parts.

Model

Serial number

Guide bar part number

Chain part number

See "Specifications" in this manual for the recommended reduced kickback cutting attachments.

Maintenance and Repairs

Users of this unit should carry out only the maintenance operations described in this manual. STIHL recommends that other repair work be performed only by authorized STIHL servicing dealers using genuine STIHL replacement parts.

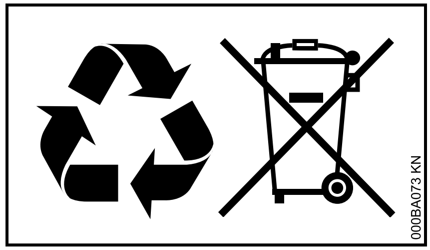
Genuine STIHL parts can be identified by the STIHL part number, the STIHL® logo and, in some cases, by the STIHL parts symbol . The symbol may appear alone on small parts.

For repairs of any component of this unit's air emissions control system, please refer to the air emissions systems warranty in this manual.

Disposal

Contact the local authorities or your STIHL servicing dealer for information on disposal.

Improper disposal can be harmful to health and pollute the environment.



- Take STIHL products including packaging to a suitable collection point for recycling in accordance with local regulations.
- Do not dispose with domestic waste.

Limited Warranty

STIHL Incorporated Limited Warranty Policy for Non-Emission-Related Parts and Components

This product is sold subject to the STIHL Incorporated Limited Warranty Policy, available at

www.stihlusa.com/warranty.html

It can also be obtained from your authorized STIHL dealer or by calling 1-800-GO-STIHL (1-800-467-8445).

A separate emissions control system warranty is provided for emission-related components.

STIHL Incorporated Federal Emission Control Warranty Statement

Your Warranty Rights and Obligations

The U.S. Environmental Protection Agency (EPA) and STIHL Incorporated are pleased to explain the Emission Control System Warranty on your equipment type engine. In the U.S. new 1997 and later model year small off-road equipment engines must be designed, built and equipped, at the time of sale, to meet the U.S. EPA regulations for small non road engines. The equipment engine must be free from defects in materials and workmanship which cause it to fail to conform with U.S. EPA standards for the first two years of engine use from the date of sale to the ultimate purchaser.

STIHL Incorporated must warrant the emission control system on your small off-road engine for the period of time listed below provided there has been no abuse, neglect or improper maintenance of your small off-road equipment engine.

Your emission control system includes parts such as the carburetor and the ignition system. Also included may be hoses, and connectors and other emission-related assemblies.

Where a warrantable condition exists, STIHL Incorporated will repair your small off-road equipment engine at no cost to you including diagnosis (if the diagnostic work is performed at an authorized dealer), parts and labor.

Manufacturer's Warranty Coverage

In the U.S., 1997 and later model year small off-road equipment engines are warranted for two years. If any emission-related part on your engine is defective, the part will be repaired or replaced by STIHL Incorporated free of charge.

Owner's Warranty Responsibilities

As the small off-road equipment engine owner, you are responsible for the performance of the required maintenance listed in your instruction manual. STIHL Incorporated recommends that you retain all receipts covering maintenance on your small off-road equipment engine, but STIHL Incorporated cannot deny warranty solely for the lack of receipts or for your failure to ensure the performance of all scheduled maintenance.

Any replacement part or service that is equivalent in performance and durability may be used in non-warranty maintenance or repairs, and shall not reduce the warranty obligations of the engine manufacturer.

As the small off-road equipment engine owner, you should be aware, however, that STIHL Incorporated may deny you warranty coverage if your small off-road equipment engine or a part has failed due to abuse, neglect, improper maintenance or unapproved modifications.

You are responsible for presenting your small off-road equipment engine to a STIHL service center as soon as a problem exists. The warranty repairs will be completed in a reasonable amount of time, not to exceed 30 days.

If you have any questions regarding your warranty rights and responsibilities, please contact a STIHL customer service representative at 1-800-467-8445 or you can write to

STIHL Inc.
536 Viking Drive, P.O. Box 2015
Virginia Beach, VA 23450-2015

www.stihlusa.com

Coverage by STIHL Incorporated

STIHL Incorporated warrants to the ultimate purchaser and each subsequent purchaser that your small off-road equipment engine will be designed, built and equipped, at the time of sale, to meet all applicable emissions regulations. STIHL Incorporated also warrants to the initial purchaser and each subsequent purchaser that your engine is free from defects in materials and workmanship which cause the engine to fail to conform with applicable emissions regulations for a period of two years.

Warranty Period

The warranty period will begin on the date the utility equipment engine is purchased by the initial purchaser. Product registration is recommended, so that STIHL has a means to contact you if there ever is a need to communicate repair or recall information about your product, but it is not required in order to obtain warranty service.

If any emission-related part on your engine is defective, the part will be replaced by STIHL Incorporated at no cost to the owner. Any warranted part which is not scheduled for replacement as required maintenance, or which is scheduled only for regular inspection to the effect of "repair or replace as necessary" will be warranted for the warranty period. Any warranted part which is scheduled for replacement as required maintenance will be warranted for the period of time up to the first scheduled replacement point for that part.

Diagnosis

You, as the owner, shall not be charged for diagnostic labor which leads to the determination that a warranted emissions part is defective. However, if you claim warranty for an emissions component and the machine is tested as non-defective, STIHL Incorporated will charge you for the cost of the emission test. Mechanical diagnostic work will be performed at an authorized STIHL servicing dealer. Emission test may be performed either at STIHL Incorporated or at any independent test laboratory.

Warranty Work

STIHL Incorporated shall remedy warranty defects at any authorized STIHL servicing dealer or warranty station. Any such work shall be free of charge to the owner if it is determined that a warranted part is defective.

Any manufacturer-approved or equivalent replacement part may be used for any warranty maintenance or repairs on emission-related parts and must be provided without charge to the owner. STIHL Incorporated is liable for damages to other engine components caused by the failure of an emissions warranted part still under warranty.

The following list specifically defines the emission-related warranted parts:

– Air Filter
– Carburetor (if applicable)
– Fuel Pump
– Choke (Cold Start Enrichment System) (if applicable)
– Control Linkages
– Intake Manifold
– Magneto or Electronic Ignition System (Ignition Module or Electronic Control Unit)
– Fly Wheel
– Spark Plug
– Injection Valve (if applicable)
– Injection Pump (if applicable)
– Throttle Housing (if applicable)
– Cylinder
– Muffler
– Catalytic Converter (if applicable)
– Fuel Tank
– Fuel Cap
– Fuel Line
– Fuel Line Fittings

- Clamps
- Fasteners

Where to Make a Claim for Warranty Service

Bring the product to any authorized STIHL servicing dealer.

Maintenance Requirements

The maintenance instructions in this manual are based on the application of the recommended 2-stroke fuel-oil mixture (see also instruction "Fuel"). Deviations from this recommendation regarding quality and mixing ratio of fuel and oil may require shorter maintenance intervals.

Limitations

This Emission Control Systems Warranty shall not cover any of the following:

- repair or replacement required because of misuse, neglect or lack of required maintenance,
- repairs improperly performed or replacements not conforming to STIHL Incorporated specifications that adversely affect performance and/or durability, and alterations or modifications not recommended or approved in writing by STIHL Incorporated,

and

- replacement of parts and other services and adjustments necessary for required maintenance at and after the first scheduled replacement point.

Trademarks

STIHL Registered Trademarks

STIHL®

The color combination orange-grey (U.S. Registrations #2,821,860; #3,010,057, #3,010,058, #3,400,477; and #3,400,476)

AutoCut®

FARM BOSS®

iCademy®

MAGNUM®

MasterWrench Service®

MotoMix®

OILOMATIC®

ROCK BOSS®

STIHL Cutquik®

STIHL DUROMATIC®

STIHL Quickstop®

STIHL ROLLOMATIC®

STIHL WOOD BOSS®

TIMBERSPORTS®

WOOD BOSS®

YARD BOSS®

Some of STIHL's Common Law Trademarks

STIHL Injection™

Team STIHL™

4-MIX ™

BioPlus ™

Easy2Start ™

EasySpool ™

ElastoStart ™

Ematic ™

FixCut ™

IntelliCarb ™

Master Control Lever ™

Micro ™

Pro Mark ™

Quiet Line ™

STIHL M-Tronic ™

STIHL OUTFITTERS ™

STIHL PICCO ™

STIHL PolyCut ™

STIHL PowerSweep ™

STIHL Precision Series ™

STIHL RAPID ™

STIHL SuperCut ™

TapAction ™

TrimCut ™

This listing of trademarks is subject to change.

Any unauthorized use of these trademarks without the express written consent of
ANDREAS STIHL AG & Co. KG, Waiblingen is strictly prohibited.

WARNING

This product contains chemicals known to the State of California to cause cancer, birth defects or other reproductive harm.

ADVERTENCIA

Este producto contiene sustancias químicas consideradas por el Estado de California como causantes de cáncer, defectos de nacimiento u otra toxicidad reproductora.

WARNING

The engine exhaust from this product contains chemicals known to the State of California to cause cancer, birth defects or other reproductive harm.

ADVERTENCIA

El gas de escape del motor de esta máquina contiene productos químicos que en el estado de California son considerados como causantes de cáncer, defectos de nacimiento u otros efectos nocivos para los órganos de la reproducción.

0458-208-8621-C

englisch / spanisch USA

USA

www.stihl.com

0458-208-8621-C

2025.03 chainsaw - Murdochs, manual attachment

STIHL®

(USA) Bar and Chain Information

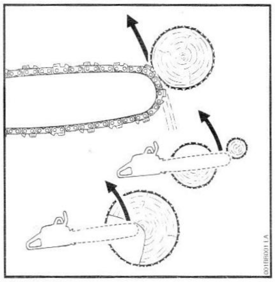
www.stihl.com

Instructions, Warnings and Directions to STIHL customers for compliance with the ANSI/OPEI B175.1-2012 Chain Saw Standard

0457 531 3023 D K19

WARNING

Because a chain saw is a high-speed wood-cutting tool, some special safety precautions must be observed as with any other power saw to reduce the risk of personal injury. Careless or improper use may cause serious or even fatal injury.



WARNING

Kickback may occur when the moving saw chain near the upper quadrant of the bar nose contacts a solid object or is pinched. When this occurs, the energy driving the saw chain can create a force that moves the chain saw in a direction opposite to the saw chain movement at the point where the saw chain is slowed or stopped. This may fling the bar up and back in a lightning fast reaction mainly in the plane of the bar and can cause severe or fatal injury to the operator.
Kickback may occur, for example, when the saw chain near the upper quadrant of the bar nose contacts the wood or is pinched during limbing or when it is incorrectly used to begin a plunge or boring cut. The greater the force of the kickback reaction, the more difficult it becomes for the operator to control the chain saw. Many factors influence the occurrence and force of the kickback reaction. These include saw chain speed, the speed at which the bar and saw chain contact the object, the angle of contact, the condition of the saw chain and other factors. The type of bar and saw chain you use is an important factor in the occurrence and force of the kickback reaction.

1. ANSI/OPEI B175.1-2012 Chain Saw Standard

Section 5.11 of ANSI/OPEI B175.1-2012 sets certain performance and design criteria related to chain saw kickback.

Section 19.108 of UL 60745-2-13 sets similar performance and design criteria for electric chain saws.

To comply with ANSI/OPEI B175.1-2012,

A) Saws with a displacement of less than 3.8 cubic inches (62 cm³)
— must, in their original condition, meet a 45° computer derived kickback angle or the moving chain shall be stopped before the chain saw exceeds the 45° computed kickback angle,
— and must be equipped with at least two devices to reduce the risk of kickback injury, such as a chain brake, low kickback chain, reduced kickback bar, etc.
B) Saws with a displacement of 3.8 cubic inches (62 cm³) and above
— must be equipped with at least one device designed to reduce the risk of kickback injury, such as a chain brake, low kickback chain, reduced kickback bar, etc.

The computer derived angles for chain saws below 3.8 cubic inches (62 cm³) displacement are measured by applying a computer program to test results from a kickback test machine.

WARNING

The computer derived angles of the chain saw standard may bear no relationship to actual kickback bar rotation angles that may occur in real life cutting situations. Compliance with the chain saw standard does not automatically mean that in a real life kickback the bar and chain will rotate at most 45°. In addition, features designed to reduce kickback injuries may lose some of their effectiveness when they are no longer in their original condition, especially if they have been improperly maintained.

WARNING

In order for chain saws below 3.8 cubic inches (62 cm³) displacement to comply with the computed kickback angle requirements of the chain saw standard use only the following cutting attachments:
— **bar and chain combinations listed as complying in the "Specifications" section of the instruction manual, the current STIHL Bar & Chain Catalog or**
— **other replacement bar and chain combinations marked in accordance with the standard for use on the chain saw or**
— **replacement chain designated "low kickback saw chain."**[1)]

WARNING

Use of other, non-listed bar / chain combinations may increase kickback forces and the risk of kickback injury. New bar / chain combinations may be developed after publication of this literature, which will, in combination with certain chain saws, comply with the chain saw standard as well. Check with your STIHL dealer for such combinations.

Please ask your STIHL dealer to properly match your chain saw with the appropriate bar / chain combination to reduce the risk of kickback injury.

If you are unsure which chain pitch and gauge and/or which guide bar lengths match your chain saw powerhead (and sprocket), ask your STIHL dealer or consult the current STIHL Bar & Chain Catalog.

2. Devices for Reducing the Risk of Kickback Injury

STIHL offers a variety of bars and chains. STIHL reduced kickback bars and low kickback chains are designed to reduce the risk of kickback injury. Other chains are designed to achieve higher cutting performance or sharpening ease, but in turn are more prone to kickback.

STIHL has developed a color code system to help you identify the STIHL reduced kickback bars and low kickback chains. Cutting attachments with green markings or green labels on the packaging are designed to reduce the risk of kickback injury. The matching of green marked chain saws under 3.8 cubic inches (62 cm³) displacement with green labeled bars and green labeled chains gives compliance with the computed kickback angle requirements of the chain saw standard when the products are in their original condition. Cutting attachments with yellow markings or labels are for users with extraordinary cutting needs, having experience and specialized training for dealing with kickback.

STIHL recommends the use of its green labeled reduced kickback bars, green labeled low kickback chains and a chain saw equipped with a STIHL Quickstop chain brake for both experienced and inexperienced chain saw users.

WARNING

Reduced kickback bars and low kickback chains do not prevent kickback, but they are designed to reduce the risk of kickback injury. They are available from your STIHL dealer.

WARNING

Even if your saw is equipped with a reduced kickback bar and / or low kickback chain, this does not eliminate the risk of injury by kickback. Therefore, always observe all safety precautions to avoid kickback situations.

[1)] A "low-kickback saw chain" is a chain that has met the kickback performance requirements of ANSI/OPEI B175.1-2012 when tested according to the provisions specified in § 9 of ANSI/OPEI B175.1-2012.

2.1 Low kickback chain

Some types of saw chains have specially designed components to reduce the force of nose contact kickback. STIHL has developed low kickback chain for your powerhead.

WARNING

STIHL saw chains should be used with kickback reducing devices such as a properly functioning chain brake and a reduced kickback bar.

WARNING

A blunt or incorrectly sharpened chain may reduce or negate the effects of the design features intended to reduce kickback energy. Improper lowering or sharpening of the depth gauges as well as changing the shape of the cutters may increase the risk and the energy of kickback. Always cut with a properly sharpened chain.

2.2 Reduced kickback bars

STIHL green labeled reduced kickback bars are designed to reduce the risk of kickback injury when used with STIHL green labeled low kickback chains.

WARNING

When used with other, more aggressive chains, these bars may be less effective in reducing kickback.

2.3 STIHL Quickstop® chain brake

STIHL has developed a chain stopping system designed to reduce the risk of injury in certain kickback situations. It is called a Quickstop chain brake. The STIHL Quickstop chain brake is available as standard equipment on newer STIHL chain saws. STIHL recommends the use of the STIHL Quickstop chain brake on your powerhead with green labeled reduced kickback bars and low-kickback chains.

WARNING

No STIHL Quickstop chain brake or other chain brake device prevents kickback. These devices are designed to reduce the risk of kickback injury, if activated, in certain kickback situations.

WARNING

An improperly maintained chain brake may increase the time needed to stop the chain after activation, or may not activate at all.

WARNING

Even if your saw is equipped with a STIHL Quickstop chain brake, a reduced kickback bar and low-kickback chain, this does not eliminate the risk of injury by kickback. Therefore, always observe all safety precautions to avoid kickback situations.

3. Maintenance Instructions for STIHL Oilomatic® Saw Chains

In order to increase the service life of the saw chain, sprocket and guide bar, always use two new chains in rotation on the new sprocket.

Tension the new chain correctly and run it without load for about 3 minutes to break it in. Ample lubrication is essential during this period. If your saw is equipped with an oil flow control, set it to maximum flow for the first hour of operation. Use only STIHL chain oil. After the break-in period, check chain tension with the engine switched off and retension as necessary.

Correct chain tension and lubrication are critically important for both the cutting results and the service life of the cutting attachment. Chain lubrication must, therefore, be checked before starting work. Chain tension should be visually checked during cutting and corrected as necessary.

WARNING

The bar and chain heat up during operation. Check and adjust chain tension while cool only. A heated up chain will sag, then tighten up as it cools. Wear gloves during adjustments.

Turn over guide bar every time you sharpen or change the chain. Clean the guide bar groove and the oil inlet hole at regular intervals.

Never work with a dull chain. Always sharpen the chain in a timely manner according to instructions. Use only STIHL sharpening aids and follow the detailed sharpening instruction in your Instruction Manual or the instructions supplied with the sharpener.

Always observe the safety precautions and warnings given in your Instruction Manual when operating the saw or sharpening the saw chain.

2025.03

Serial ID's related to
Stihl MS250 from
Murdochs →

↳ original box not provided
with sale

3005 008 7017	AB	
< 45 cm/ 18"		t=a:2
200, 230, 231, 241, 250, 251	7	68	8,25 mm/ .325"

7 95711 35812 9

1,3 mm
0.050"

0000 967 4913 A D 20

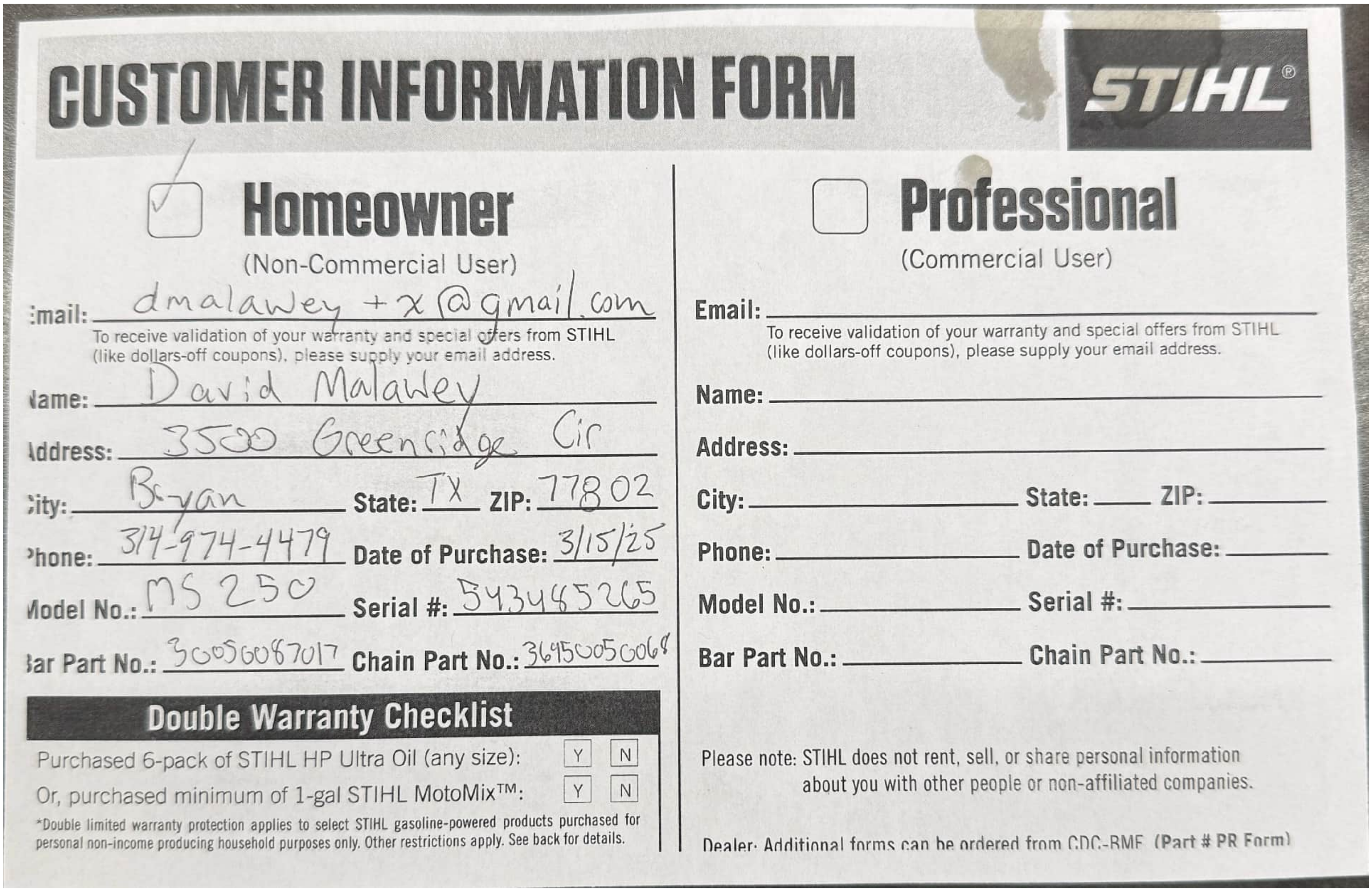
MICRO	8,25mm .325"	1,3mm .050"	68	<40cm 16" <45cm 18" STIHL
4,8mm 3/16"	0,65mm 0.025"	30 °	3695 005 0068	
			23 RM3	68

MICRO	8,25mm .325"	1,3mm .050"	68	<40cm 16" <45cm 18" STIHL
3695 005 0068	8 86661 88094 2			
23 RM3 68				


STIHL®
1123/06 B
A C C
1123-011-3078
591
MS 250-Z .325" Chainsaw
543485265
Manufactured by STIHL INC.,USA.
with domestic and foreign components
Fabrique par STIHL Inc. Etats-Unis, avec
des composants domestiques et etrangers

Customer Information Form, Stihl Chainsaw service
from College Station Murdoch's local store.

(photograph on 3/15/2025)



CUSTOMER INFORMATION FORM

STIHL®

☑ **Homeowner**
(Non-Commercial User)

:mail: dmalawey+x@gmail.com
To receive validation of your warranty and special offers from STIHL (like dollars-off coupons), please supply your email address.

lame: David Malawey

ddress: 3500 Greenridge Cir

ity: Bryan **State:** TX **ZIP:** 77802

hone: 314-974-4479 **Date of Purchase:** 3/15/25

lodel No.: MS 250 **Serial #:** 543445265

ar Part No.: 3005008 7017 **Chain Part No.:** 36450050068

Double Warranty Checklist

Purchased 6-pack of STIHL HP Ultra Oil (any size):	Y	N
Or, purchased minimum of 1-gal STIHL MotoMix™:	Y	N

*Double limited warranty protection applies to select STIHL gasoline-powered products purchased for personal non-income producing household purposes only. Other restrictions apply. See back for details.

☐ **Professional**
(Commercial User)

Email:
To receive validation of your warranty and special offers from STIHL (like dollars-off coupons), please supply your email address.

Name:

Address:

City: **State:** **ZIP:**

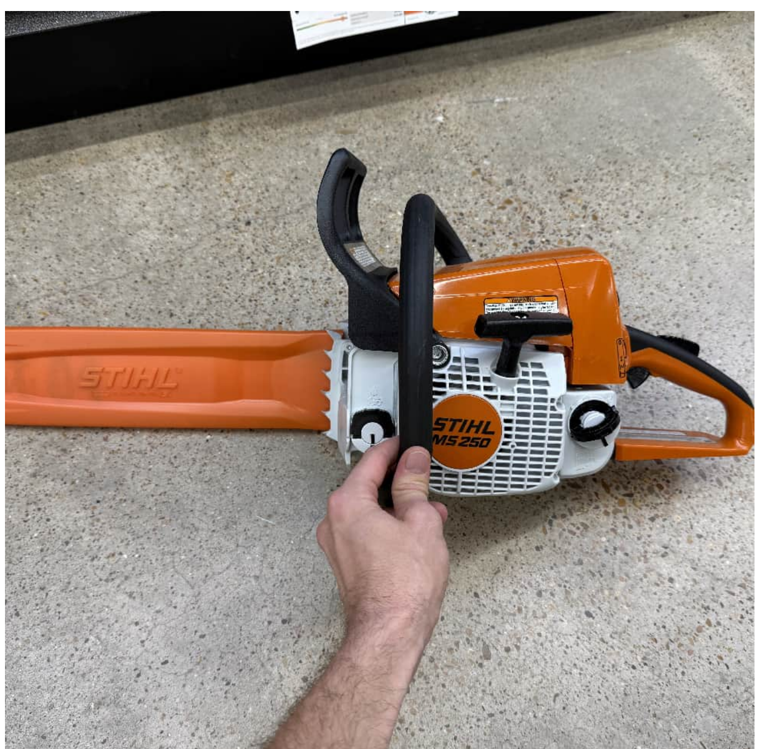
Phone: **Date of Purchase:**

Model No.: **Serial #:**

Bar Part No.: **Chain Part No.:**

Please note: STIHL does not rent, sell, or share personal information about you with other people or non-affiliated companies.

Dealer: Additional forms can be ordered from CDC-BMF (Part # PR Form)



Stihl Chainsaw - MS 250 - Murdoch's, Bryan, TX

2025.03.15

MURDOCH'S

RANCH & HOME SUPPLY

murdochs.com

Murdoch's Ranch & Home Supply
Murdoch's in College Station
1502 Harvey Rd 200
College Station TX 77840
Phone:(979) 977-4672

Chainsaw, Stihl MS250
18-in bar

3/15/2025 03:27 PM Term:CLG-POS-06 Clerk:ALMH
Trans #: 54-CLG-POS-06-1742070313460

Item	Description	Qty	UOM	Price	Total
1155373		1	EA	$399.99	$399.99
STIHL MS 250 Z 18in					
				Net Price:	$399.99

david malawey
3500 greenridge cir bryan 77802
314-974-4479
543485265

Subtotal:	$399.99
Tax:	$33.00
Total:	$432.99
CREDIT/DEBIT	$432.99

CARD TYPE	: CREDIT CARD
TRANSACTION	: Sale
TRANSACTION ID	: 54069360
DATE/TIME	: 3/15/2025 3:27:21 PM
CLERK ID	: ALMH
CARD	: Visa
CARD NUMBER	: 9091
NAME	: MALAWEY/DAVID
INVOICE	: A0000000031010
AUTH CODE	: 01580D
REF.NO	: 40
ENTRY MODE	: Chip
AMOUNT	: $432.99
STATUS	: Approved

5 4 0 6 0 3 1 5 2 5 0 3 9 3

Thank you for shopping at your local
Murdoch's Ranch & Home Supply

For warranty/return/layaway questions visit:
Murdochs.com/customer-service/

Serial Number - Photograph 2025.03.17

serial: 1123-011-3078
SKU: 543485265

Product Tags - package, specs, box, photograph

Note

Chain Spec

Bar length: 45cm
Bar thickness: 1.3mm
Tooth Pitch: 8.25mm
No. Links: 68
Sprocket Teeth: 7

Chain Spec
Style: micro